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Digital twin technology and operations management efficiency in construction: the roles of information transparency, collaboration and digital readiness

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Abstract

Purpose – This study examines how Digital Twin Technology (DTT), defined as a virtual replica of physical systems enabled by real-time data, influences operations management efficiency (OME) in the construction industry. Drawing on Resource Orchestration Theory (ROT), it explores the organizational mechanisms through which DTT contributes to OME, focusing on information transparency (IT) and collaborative decision-making (CDM), and examines organizational digital readiness (ODR) as a contextual condition shaping the DTT–OME relationship.

Design/methodology/approach – The study adopts a quantitative research design using survey data collected from 313 construction professionals. Data were analyzed using partial least squares structural equation modeling to test the hypothesized relationships, and necessary condition analysis was used to assess critical enabling factors. The model evaluates direct, mediating, and moderating effects among DTT, IT, CDM, OME, and ODR.

Findings – Results reveal that DTT has a significant positive effect on OME and strongly enhances both IT and CDM. IT exerts the greatest influence on OME, while CDM contributes as a complementary factor. Both IT and CDM mediate the relationship between DTT and OME, indicating that operational improvements occur through enhanced information visibility and coordinated decision-making. ODR significantly moderates the DTT–OME relationship in a negative direction. Although this finding is contrary to the hypothesized positive moderation, the effect suggests that the marginal operational contribution of DTT may diminish as baseline digital readiness increases, potentially reflecting overlapping digital capabilities, integration complexity, and established organizational routines.

Originality/value – This study examines how DTT improves operational efficiency in construction by enhancing IT and CDM. Rather than treating DTT adoption as an end in itself, the study focuses on the pathway through which continuously synchronized digital information is transformed into organizational mechanisms that support construction operations. It further identifies ODR as a boundary condition that shapes the marginal value of DTT. By focusing on fragmented information flows, temporary organizational structures, and multi-stakeholder coordination, the study provides a construction-specific explanation of digital twin value creation.

Keywords Digital twin technology, Operations management efficiency, Information transparency, Collaborative decision-making, Organizational digital readiness

Paper type Research article

1. Introduction

The construction industry is experiencing rapid digital transformation driven by emerging technologies associated with Industry 4.0. Among these technologies, Digital Twin Technology (DTT) has gained significant attention due to its ability to create virtual

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representations of physical assets, systems, and processes. Digital twins integrate real-time data, simulation, and monitoring capabilities, allowing organizations to visualize project activities, predict operational issues, and improve decision-making (Zhou *et al.*, 2020). In complex construction environments where projects involve multiple stakeholders and dynamic operations, digital twins can enhance coordination, optimize resource allocation, and improve operational control. Consequently, organizations are increasingly adopting DTT to improve operational efficiency and overall project performance (Singh *et al.*, 2026). The emergence of Industry 4.0 has accelerated the adoption of cyber-physical systems, Internet of Things (IoT), cloud computing, and advanced analytics within construction processes, creating the technological foundation for digital twin applications (Oesterreich and Teuteberg, 2016; Sawhney *et al.*, 2020).

Unlike manufacturing environments characterized by fixed production lines, standardized workflows, and stable organizational structures, construction projects operate within highly dynamic and continuously changing physical environments. Construction sites evolve daily as project stages progress, creating what may be described as a form of spatiotemporal fluidity in which resources, personnel, equipment, and work conditions are constantly reconfigured. Furthermore, construction projects rely on temporary coalitions of contractors, subcontractors, consultants, suppliers, and owners, many of whom maintain separate information systems and are often reluctant to share proprietary data. Consequently, understanding whether and how DTT creates operational value in construction requires attention to construction-specific mechanisms rather than assuming direct generalization from other industries.

Despite these potential benefits, many construction firms still face challenges in translating digital technology adoption into improved operations management efficiency (OME) (Tetik *et al.*, 2019). Construction projects often suffer from fragmented information systems, limited transparency, and weak coordination among stakeholders, which reduce operational effectiveness (Froese, 2010; Turk, 2016; Tetik *et al.*, 2019). These information and coordination problems are closely connected to persistent operational challenges in construction, including project delays, poor productivity, resource shortages, and difficulties in planning and decision-making. A quantitative review of construction-delay research identified execution problems, administrative problems, and labor conflicts as the three major categories of delay causes, highlighting the persistent operational difficulties encountered during construction execution (Viles *et al.*, 2020). Similarly, Zidane and Andersen (2018) identified poor planning and scheduling, slow or poor decision-making, resource shortages, and poor communication and coordination between project participants among the recurring factors contributing to construction delays. The operational consequences of coordination failures are also observable at the activity level: Hwang *et al.* (2014) found that trade-level coordination problems in construction were associated with substantial rework and waiting time, demonstrating how coordination deficiencies can translate into measurable productivity losses.

These longstanding challenges have been widely documented in construction management and construction informatics research, where fragmented project information, poor interoperability, and delayed communication frequently lead to schedule overruns, resource conflicts, rework, and reduced project productivity (Hwang *et al.*, 2014; Hegazy *et al.*, 2011). These observations are consistent with earlier construction informatics research demonstrating that effective information integration and interoperability are fundamental prerequisites for improving collaboration and project performance in construction (Nitithamyong and Skibniewski, 2004; Succar, 2009). Consequently, the critical question is no longer whether construction organizations should adopt digital technologies, but rather how technologies capable of continuously connecting physical project conditions with digital information can address these operational challenges and improve the execution of construction activities. To better understand this relationship, it is important to examine the organizational mechanisms through which DTT creates value.

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In this study, information transparency (IT) refers to the availability and accessibility of accurate and real-time project information (Duong *et al.*, 2025), while collaborative decision-making (CDM) refers to the joint participation of stakeholders in planning, coordination, and problem-solving activities (Wang *et al.*, 2025). In construction projects, IT manifests as real-time visibility into site progress, equipment utilization, safety conditions, material movements, and schedule performance across multiple project participants. Similarly, CDM occurs when project managers, contractors, engineers, and subcontractors collectively evaluate project conditions and coordinate operational responses using shared digital information. This operational role of CDM is particularly important in construction because coordination failures among trades can generate waiting and rework and consequently reduce productivity (Hwang *et al.*, 2014). The effectiveness of digital technologies may depend on an organization's ability to adopt and utilize digital tools. Organizational digital readiness (ODR) reflects the technological infrastructure, employee competencies, and managerial support required to successfully implement digital transformation initiatives (Alotaibi *et al.*, 2025). Given the temporary and decentralized nature of construction organizations, ODR extends beyond technical readiness and includes the capability to coordinate digital processes across multiple independent project actors.

DTT is closely related to, but conceptually distinct from, Building Information Modeling (BIM). BIM has become the dominant digital information management paradigm in construction and provides a structured repository of project information across design, construction, and operation phases (Succar, 2009; Volk *et al.*, 2014). Foundational BIM research has consistently recognized BIM as an enabler of integrated information management, multidisciplinary collaboration, and lifecycle decision support across construction projects (Succar, 2009; Gu and London, 2010; Volk *et al.*, 2014). BIM primarily provides a static or periodically updated digital representation of building information across the project lifecycle, whereas DTT extends this capability through continuous, bidirectional synchronization between physical assets and their virtual counterparts. Recent studies position digital twins as an extension of BIM through the integration of IoT-enabled sensing, real-time data streams, and continuous feedback mechanisms (Boje *et al.*, 2020; Opoku *et al.*, 2021). In practical terms, BIM generally reflects what has been designed or documented, whereas DTT reflects what is currently occurring on-site through continuous data exchange between the physical and digital environments. For example, sensor-enabled equipment, drones, wearable devices, and IoT platforms can continuously update the digital twin regarding site conditions, resource utilization, safety events, and project progress, while simulations generated within the digital twin can subsequently guide operational decisions in the physical environment (Patil *et al.*, 2024; Nyqvist *et al.*, 2024). Through sensors, IoT connectivity, and real-time analytics, DTT enables dynamic monitoring, simulation, and operational adjustment during construction execution.

Importantly, the operational value of DTT extends beyond the information management capabilities traditionally associated with BIM. While BIM provides a reliable digital representation for project planning, design coordination, and documentation, DTT enables continuous interaction between physical construction activities and digital models through bidirectional information exchange. This capability allows project teams to receive live operational information from construction sites while simultaneously using predictive simulations and analytical insights generated by the digital twin to influence subsequent on-site activities. Consequently, DTT supports proactive rather than reactive project management by enabling early detection of schedule deviations, equipment failures, material shortages, safety risks, and productivity bottlenecks before they significantly affect project performance.

Furthermore, DTT creates value by facilitating real-time coordination among contractors, subcontractors, consultants, suppliers, and project managers who frequently operate using different organizational systems and possess incomplete information. Through continuously synchronized digital environments, project participants can evaluate emerging site conditions

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simultaneously, assess alternative operational scenarios, and collectively determine appropriate responses. Such continuous coordination reduces information asymmetry, minimizes decision delays, improves cross-functional collaboration, and enhances the alignment between project planning and execution. These capabilities distinguish DTT from conventional digital technologies by transforming digital information into dynamic operational intelligence that directly supports construction execution rather than merely documenting project information.

Previous studies have investigated the application of DTT primarily in manufacturing and supply chain contexts, where operations are relatively stable and digitally integrated (Polimetla *et al.*, 2025; Guo and Mantravadi, 2025). Although these studies demonstrate the performance benefits of digital twins, they do not fully explain how DTT generates operational value in construction settings characterized by fragmented information flows, temporary project structures, and extensive stakeholder interdependence. Furthermore, existing construction informatics research emphasizes persistent challenges associated with information fragmentation and interoperability across project stakeholders, suggesting the need to investigate how DTT can overcome these long-standing limitations (Froese, 2010; Turk, 2016). These long-recognized challenges further reinforce the need for digital technologies that not only integrate project information but also facilitate collaboration and informed decision-making across multiple project participants (Nitithamyong and Skibniewski, 2004; Gu and London, 2010). Moreover, existing research largely emphasizes direct performance effects, while the organizational mechanisms through which DTT influences operations remain underexplored (Angelopoulos *et al.*, 2023; Verhoef *et al.*, 2021). Specifically, limited attention has been given to IT and CDM as mediating mechanisms, and the moderating role of ODR has received relatively little empirical investigation in construction environments (Nyqvist *et al.*, 2024; Naji *et al.*, 2024). Accordingly, the scope of this study is deliberately focused on the operational value-creation pathway of DTT rather than on DTT adoption itself. The study examines whether DTT contributes to OME and, more specifically, whether this contribution operates through IT and CDM, while ODR is examined as a boundary condition. This study addresses these gaps by examining how DTT improves OME in Chinese construction organizations through IT and CDM, and by testing whether ODR conditions the DTT–OME relationship. By focusing on construction-specific operational processes and distinguishing DTT from broader digital technologies and BIM-based approaches, the study offers a more contextually grounded explanation of digital twin value creation.

Accordingly, this study extends existing knowledge by explaining the organizational processes through which DTT creates operational value in construction projects rather than proposing an entirely new theoretical perspective. Specifically, the study integrates insights from construction informatics, BIM, Industry 4.0, and Resource Orchestration Theory (ROT) to explain how real-time digital resources are transformed into IT and CDM, thereby improving OME. By embedding DTT within construction-specific operational processes and explicitly distinguishing its dynamic capabilities from the predominantly information-centric role of BIM, the study contributes a more contextually grounded understanding of digital twin value creation in project-based construction environments.

This paper is organized as follows. Section 2 presents the theoretical background and Section 3 develops the research hypotheses and the conceptual model. Section 4 describes the research methodology, including the sample, measurement scales, and analytical techniques (PLS-SEM and NCA). Section 5 reports the empirical results. Section 6 discusses the findings, outlines theoretical and managerial implications, and Section 7 concludes with limitations and future research directions.

2. Theoretical background: Resource Orchestration Theory (ROT)

ROT emerged as an extension of the Resource-Based Theory (RBT) to address its key limitation, the lack of explanation regarding how managers actively utilize resources to

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generate value (Sirmon *et al.*, 2007). While RBT emphasizes the importance of valuable, rare, and inimitable resources, ROT focuses on the managerial role in structuring, bundling, and leveraging resources to achieve superior organizational performance and competitive advantage. ROT highlights that resources alone are insufficient unless they are effectively orchestrated through managerial actions (Iliyas and Barca, 2025). ROT is typically conceptualized through three key perspectives: *firm scope*, *hierarchical levels*, and *stage of maturity* (Sirmon *et al.*, 2011). *From the firm's scope perspective*, managers align resources with organizational strategies to create value. *From the hierarchical perspective*, effective coordination across different managerial levels is necessary to ensure proper resource utilization. *From a maturity perspective*, organizations must adapt their resource orchestration strategies according to their lifecycle stage to maintain performance and competitiveness. These perspectives collectively emphasize that resource value is realized through effective coordination and deployment rather than mere possession.

ROT has increasingly been adopted within operations management and digital transformation research because it provides a process-oriented explanation of how technological investments are converted into organizational capabilities and performance outcomes. Rather than assuming that digital technologies automatically generate competitive advantages, ROT proposes that managers must deliberately organize technological resources, integrate them with organizational routines, and coordinate complementary capabilities before performance improvements can be realized. This emphasis on managerial orchestration makes ROT particularly suitable for studying emerging digital technologies whose value depends on organizational implementation rather than technological acquisition alone.

ROT has been widely applied in operations and supply chain management research to explain how firms mobilize and coordinate technological, informational, and organizational resources to generate superior performance (Sirmon *et al.*, 2011; Chirico *et al.*, 2011; Hitt *et al.*, 2011). Prior studies have demonstrated its relevance in explaining how organizations leverage technological and organizational resources to enhance performance outcomes (Lin *et al.*, 2025). Farrukh *et al.* (2026) emphasized how firms orchestrate analytics capabilities to enhance supply chain performance. Similarly, IT competence supports flexibility through effective resource orchestration and the coordination of resources across multi-tier supply chains to support sustainability (Gligor *et al.*, 2022). These studies confirm that organizational capabilities emerge from effective resource orchestration rather than isolated resource ownership. ROT has been applied in digital transformation contexts. DTT improves operational outcomes when supported by organizational capabilities, and digital twins enhance operational performance through integration and coordination mechanisms (Gligor *et al.*, 2022). Across these studies, the common finding is that digital technologies create value through organizational processes that transform digital resources into operational capabilities rather than through the technologies themselves.

Although ROT has demonstrated considerable explanatory power within manufacturing and supply chain settings, its application to construction remains comparatively limited despite the industry's distinctive organizational characteristics. Unlike manufacturing firms, construction organizations operate through temporary, project-based networks comprising owners, contractors, subcontractors, consultants, suppliers, and regulatory agencies. These organizations continuously exchange information while simultaneously managing changing physical environments, evolving project requirements, and decentralized decision-making structures. Consequently, the effectiveness of digital technologies in construction depends not only on technological functionality but also on organizations' ability to orchestrate information, coordinate multiple stakeholders, and integrate dispersed project resources throughout the project lifecycle.

The construction literature provides additional justification for applying ROT to DTT adoption. Research on construction informatics and Industry 4.0 highlights that operational inefficiencies often stem from fragmented information systems, weak inter-organizational coordination, and limited integration of digital resources across project stages. BIM studies

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have shown that digital representations can improve information management, but their effectiveness depends on organizational capabilities that enable integration, coordination, and decision support. ROT offers a suitable theoretical lens because it explains how managers can structure and leverage digital resources, such as DTT, data infrastructures, and collaborative processes, to create operational value in complex project environments.

Construction informatics research has consistently recognized BIM as the dominant platform for managing digital project information throughout design, construction, and facility operation (Succar, 2009; Volk *et al.*, 2014). Nevertheless, BIM primarily functions as an information repository that supports project documentation, visualization, clash detection, and multidisciplinary coordination. While these capabilities significantly improve information management, they generally depend on periodic model updates and therefore provide limited support for continuous operational control during project execution. DTT extends BIM by integrating IoT sensors, cyber-physical systems, real-time monitoring, predictive analytics, and bidirectional communication between physical construction assets and their virtual counterparts (Boje *et al.*, 2020; Opoku *et al.*, 2021). Consequently, DTT should not be viewed as a replacement for BIM but rather as an operational extension that transforms static digital information into continuously evolving operational intelligence capable of supporting dynamic project execution and decision-making.

From a ROT perspective, BIM and DTT therefore perform complementary rather than identical roles. BIM provides an important digital resource by establishing standardized project information, whereas DTT orchestrates that resource through continuous sensing, simulation, feedback, and operational coordination. Managers create value not simply because digital information exists, but because they structure, integrate, and leverage continuously updated project information to improve transparency, facilitate CDM, and enhance operational efficiency. This distinction directly explains why DTT contributes additional operational value beyond conventional BIM implementations.

This perspective also aligns with research on absorptive capacity and information technology-enabled capabilities, which emphasizes that technological investments yield performance benefits only when organizations possess the capabilities required to assimilate, integrate, and exploit new knowledge. Absorptive capacity theory suggests that organizations derive value from new technologies only when they possess the capability to acquire, assimilate, transform, and exploit external knowledge (Cohen and Levinthal, 1990; Zahra and George, 2002). Similarly, IT-enabled capability research argues that digital investments create performance benefits when supported by complementary organizational processes and managerial capabilities (Bharadwaj, 2000; Wade and Hulland, 2004).

This study conceptualizes DTT as a strategic technological resource that differs from conventional digital systems because it enables bidirectional, real-time synchronization between physical construction assets and their virtual representations. Through continuous monitoring, simulation, and data integration, DTT provides dynamic operational visibility that can be leveraged through organizational processes such as IT and CDM.

Unlike conventional digital technologies that primarily capture or store project information, DTT continuously generates operational knowledge by integrating live data from equipment, sensors, drones, wearable technologies, and IoT-enabled infrastructure into digital representations of ongoing construction activities. These continuously updated representations enable managers to evaluate alternative operational scenarios, anticipate disruptions, optimize resource allocation, and coordinate responses among multiple project participants. Consequently, DTT represents a strategic digital resource whose value depends on managerial orchestration rather than technological deployment alone.

ROT is particularly appropriate for the present study because the central research question concerns not merely whether DTT improves operational outcomes, but how construction organizations transform digital resources into operational value. Alternative theoretical perspectives such as the Dynamic Capabilities View primarily emphasize organizational adaptation, Information Processing Theory focuses on information requirements and

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uncertainty reduction, and the Technology–Organization–Environment framework explains technology-adoption determinants. While each perspective offers useful insights, none explicitly explains how managers structure, combine, and deploy technological, informational, and organizational resources after adoption to generate performance outcomes. ROT directly addresses this value-creation process and therefore provides a more suitable lens for examining the mechanisms linking DTT, IT, CDM, and OME.

Within the proposed model, DTT functions as the strategic digital resource that managers acquire and structure through investments in digital infrastructure, sensing technologies, and data integration capabilities. IT reflects the structuring of information resources, as information generated by DTT must be organized, standardized, and made accessible across project stakeholders to reduce information asymmetry and improve visibility. CDM represents the bundling and leveraging of these structured information resources by integrating transparent information with stakeholder expertise, coordination routines, and collaborative processes to support collective decision-making. Together, these resource orchestration processes contribute to OME, which represents the performance outcome of successful resource orchestration. ODR functions as a boundary condition that influences the effectiveness of these orchestration processes by providing the technological infrastructure, employee competencies, and managerial commitment necessary for resource deployment.

More specifically, resource structuring occurs when managers establish the technological infrastructure required for continuous data acquisition, system integration, and digital connectivity across construction projects. Resource bundling subsequently combines these digital resources with organizational processes that improve IT by ensuring that project participants access consistent, timely, and reliable operational information. Resource leveraging occurs when transparent information is collectively interpreted by contractors, engineers, subcontractors, and project managers to support CDM and coordinated operational responses. Through this sequence, digital resources are progressively transformed into organizational capabilities that ultimately enhance OME. This explicit orchestration process provides the theoretical foundation linking DTT, IT, CDM, ODR, and OME within the proposed conceptual model.

The three ROT perspectives introduced earlier also align with the research model. From the firm-scope perspective, organizations align DTT investments with operational objectives. The hierarchical perspective, IT and CDM facilitate coordination across project managers, engineers, contractors, and subcontractors. From the maturity perspective, organizations with higher ODR are better positioned to exploit DTT capabilities and realize efficiency gains.

3. Hypotheses and research model

Based on ROT, this study develops a research model comprising six hypotheses to explain how digital resources are transformed into operational outcomes. Consistent with the resource orchestration process, DTT is conceptualized as the focal technological resource, IT represents resource structuring, CDM represents resource bundling and leveraging, and OME reflects the resulting operational performance outcome. Specifically, this study argues that the effective orchestration of DTT by managers enables organizations to develop key capabilities such as IT and CDM, which ultimately enhance OME. In this context, managers play a critical role in structuring, coordinating, and leveraging digital resources within construction projects.

Unlike conventional digital transformation studies that primarily examine technology adoption as a direct antecedent of organizational performance, this study focuses on the operational processes through which DTT creates value in project-based construction environments. The proposed model argues that the performance benefits of DTT are realized not simply because organizations possess advanced digital technologies, but because managers effectively orchestrate continuously generated digital information into organizational capabilities that improve operational execution. This perspective reflects the

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central premise of ROT that technological resources require complementary organizational mechanisms before they generate measurable operational outcomes.

The proposed model also explicitly distinguishes DTT from BIM and other digital construction technologies. While BIM establishes a shared digital representation of project information and supports design coordination, documentation, and information management throughout the project lifecycle, DTT continuously synchronizes physical construction activities with their virtual counterparts through IoT-enabled sensing, real-time monitoring, predictive simulation, and bidirectional information exchange. Consequently, DTT extends the value of BIM beyond information management by enabling dynamic operational control, continuous situational awareness, and rapid collaborative responses to changing site conditions during project execution. This distinction provides the theoretical foundation for the proposed hypotheses because the organizational mechanisms examined in this study, including IT and CDM, depend upon continuously updated operational information rather than periodically revised project documentation.

Within construction projects, these capabilities become particularly valuable because project participants frequently operate across organizational boundaries under conditions of uncertainty, evolving schedules, resource constraints, and changing site conditions. Continuous synchronization between physical operations and digital representations enables project managers, contractors, engineers, and subcontractors to identify emerging problems earlier, evaluate operational alternatives collectively, and implement coordinated responses before disruptions significantly affect project performance. Accordingly, the proposed research model explains how DTT creates operational value through construction-specific organizational mechanisms rather than assuming that performance improvements arise directly from technology adoption.

Table 1 presents the constructs and their definitions, and Figure 1 presents the research model of this study.

3.1 Digital twin technology and operations management efficiency

Effective management of construction operations is essential for improving project performance in complex and dynamic environments. In this context, DTT provides a powerful mechanism for enhancing operational control by enabling real-time monitoring, simulation, and data integration across project activities (Mohanraj and Balaji, 2025). Through these capabilities, DTT allows organizations to detect inefficiencies, anticipate potential disruptions, and adjust operational processes on time. These capabilities are consistent with Industry 4.0 objectives of integrating cyber-physical systems and data-driven decision-making to improve operational performance and process efficiency (Oesterreich and Teuteberg, 2016; Sawhney *et al.*, 2020). Since construction projects often involve fragmented workflows and multiple stakeholders, the use of digital twins improves process visibility and coordination, leading to more stable and controlled operations (Afrazi *et al.*, 2025).

In construction settings, these benefits are particularly important because operational activities occur across constantly changing physical environments where schedule deviations, equipment conflicts, material shortages, and safety incidents can emerge unexpectedly. By continuously synchronizing site conditions with digital representations, DTT enables project teams to identify emerging problems earlier and implement corrective actions before they escalate into significant operational disruptions.

Unlike conventional project information systems that primarily record or communicate project information, DTT continuously links physical construction activities with their virtual representations through bidirectional information exchange. This capability enables construction managers to monitor site conditions as they evolve, evaluate predictive simulations based on current operational data, and implement corrective actions before project disruptions become critical. Consequently, DTT supports proactive rather than reactive

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Table 1. Constructs and definitions

Construct	Definition	Adapted from
Digital twin technology (DTT)	DTT refers to a continuously synchronized digital representation of physical assets, processes, and construction activities that enables real-time monitoring, bidirectional data exchange, simulation, prediction, and operational optimization. Unlike conventional digital models, DTT continuously integrates live information from physical environments to support dynamic operational decision-making throughout project execution	Arowoiya *et al.* (2026)
Information transparency (IT)	Within construction projects, information transparency reflects stakeholders' ability to access accurate, timely, and continuously updated operational information, thereby reducing information asymmetry and supporting coordinated project execution	Yoon and Pishdad-Bozorgi (2022)
Collaborative decision-making (CDM)	CDM refers to the extent to which contractors, subcontractors, engineers, consultants, and project managers jointly evaluate project conditions, coordinate operational activities, and make collective decisions using shared digital information	Tang *et al.* (2025)
Organizational digital readiness (ODR)	ODR is the extent to which an organization possesses the necessary digital infrastructure, skills, culture, and strategic alignment to adopt and effectively utilize digital technologies. It reflects the organization's preparedness for digital transformation	Koh *et al.* (2026)
Operations management efficiency (OME)	OME refers to the ability of an organization to optimize its processes, resources, and activities to achieve higher productivity, reduced costs, and improved performance outcomes	Enad and Gerinda (2022)

Source(s): Authors' own creation/work

operations management by transforming continuously generated site information into timely operational decisions.

This operational capability also distinguishes DTT from BIM. Although BIM provides an integrated digital model that supports planning, visualization, and multidisciplinary coordination, its primary function is to manage project information rather than continuously monitor live construction operations. DTT extends BIM by integrating IoT devices, sensors, drones, wearable technologies, and cyber-physical systems that continuously update the digital model with current site conditions while simultaneously enabling predictive analyses that influence subsequent construction activities. Through this bidirectional synchronization, DTT provides construction managers with dynamic operational intelligence rather than static project documentation, thereby improving responsiveness to changing site conditions.

From the ROT perspective, DTT represents a strategic digital resource whose value depends on managerial orchestration rather than technological ownership alone. Managers must structure digital infrastructures capable of collecting real-time operational data, integrate these data across multiple project participants, and leverage continuously updated information to coordinate operational activities effectively. As a result, DTT contributes to operational efficiency because it enables managers to orchestrate information, people, and resources more effectively throughout project execution.

OME reflects the ability of organizations to execute project activities effectively while minimizing delays, reducing waste, and optimizing resource utilization. Recent studies highlight that digital technologies significantly improve operational efficiency by enhancing

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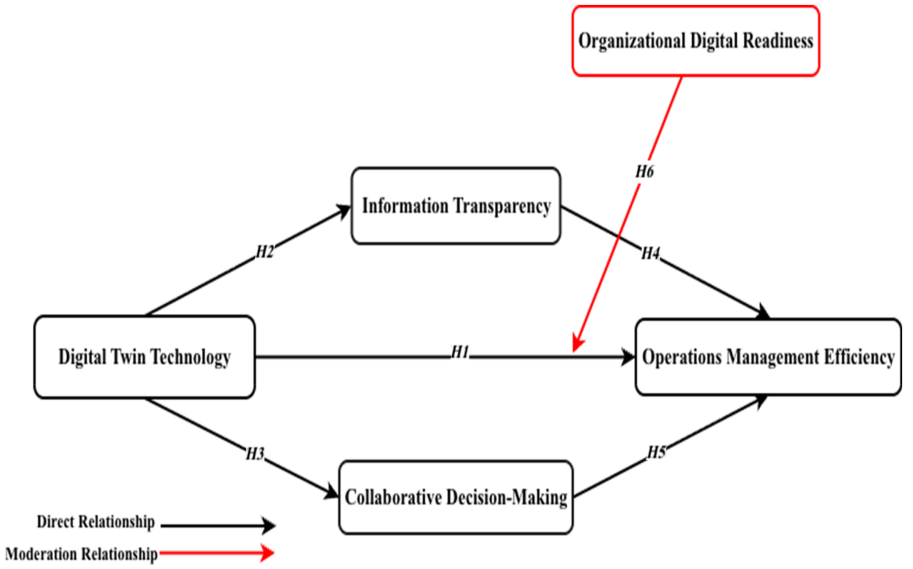


Figure 1. Research model. **Source(s):** Authors' own creation/work

process integration, real-time responsiveness, and decision-making quality (Xue *et al.*, 2025; Tian *et al.*, 2023). In particular, DTT supports continuous performance monitoring, improves scheduling accuracy, and enables better alignment between planning and execution (Mithas *et al.*, 2022). These capabilities help organizations reduce rework, improve workflow consistency, and ensure more efficient use of resources under complex project conditions.

Within project-based construction environments, these improvements extend beyond internal operational efficiency. Real-time synchronization enables contractors, subcontractors, engineers, and project managers to coordinate work using a common operational picture of site conditions, thereby reducing information asymmetry, minimizing resource conflicts, improving schedule adherence, and accelerating responses to unforeseen events. Continuous monitoring also facilitates predictive maintenance of construction equipment, proactive management of safety risks, optimization of material deliveries, and dynamic adjustment of work schedules, all of which contribute to higher levels of OME.

Furthermore, because construction projects involve temporary organizational structures and frequent interactions among independent stakeholders, operational efficiency depends heavily on the ability to coordinate decisions using accurate and continuously updated information. DTT provides this capability by integrating heterogeneous project data into a shared operational environment that supports synchronized planning and execution across organizational boundaries. Consequently, organizations adopting DTT are better positioned to improve workflow stability, reduce operational uncertainty, and achieve superior operational performance compared with organizations relying solely on conventional BIM-based information management systems.

Accordingly, the expected contribution of DTT to OME derives not merely from the information-management capabilities already associated with BIM, but from DTT's continuous synchronization, predictive simulation, and bidirectional feedback capabilities that support operational control during construction execution. Thus, we hypothesize that:

H1. DTT has a positive effect on OME.

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3.2 Digital twin technology and information transparency

IT plays a critical role in construction projects, where multiple stakeholders depend on accurate and timely information to coordinate activities and make decisions (Kor *et al.*, 2023). However, construction environments are often characterized by fragmented data, limited visibility, and delays in information sharing, which reduce project efficiency and increase the risk of errors (Zha *et al.*, 2023). Such errors may include inaccurate progress reporting, duplicated resource allocations, delayed identification of safety hazards, inconsistent design interpretations, and schedule conflicts among subcontractors. These problems often arise because stakeholders rely on disconnected information sources and lack access to a common, real-time view of project conditions. In this context, DTT provides a structured and integrated platform that enables real-time data collection, visualization, and sharing across project stages (Omrany *et al.*, 2023). By creating a digital representation of physical processes, DTT allows stakeholders to access consistent and up-to-date information, improving overall transparency within the project environment (Teisserenc and Sepasgozar, 2021).

Construction management research has long identified information fragmentation as one of the principal barriers to effective project delivery because project participants frequently operate using different software platforms, organizational procedures, and reporting structures (Fellows and Liu, 2012). Although BIM has substantially improved the management and exchange of project information by providing a shared digital representation of project data, the information contained within BIM models is often updated periodically and may not fully reflect rapidly changing on-site conditions during project execution (Papadonikolaki *et al.*, 2019). Consequently, construction stakeholders may continue to experience delays in obtaining accurate operational information when unexpected events occur on-site.

DTT extends the capabilities of BIM by continuously synchronizing physical construction activities with digital models through IoT sensors, drones, wearable technologies, equipment monitoring systems, and other cyber-physical technologies (Arowoiya *et al.*, 2026). Rather than relying solely on manually updated project information, DTT continuously captures operational data relating to construction progress, equipment utilization, workforce activities, material movements, environmental conditions, and safety performance. This continuous bidirectional exchange ensures that digital representations remain aligned with actual site conditions, thereby providing stakeholders with a common and continuously updated operational picture of the project.

The use of DTT enhances IT by reducing information asymmetry and improving data accessibility among project participants (Chen *et al.*, 2023). It enables continuous tracking of project progress, supports accurate reporting, and facilitates better communication across different organizational levels. Recent studies indicate that digital technologies significantly improve transparency by enhancing data visibility, integration, and real-time information exchange (Xue *et al.*, 2025; He *et al.*, 2025). In particular, DTT provides a centralized and reliable source of information, which helps reduce misunderstandings, improve coordination, and support informed decision-making in complex operational settings.

From a ROT perspective, DTT represents a digital resource that must be effectively structured before it can improve organizational performance. Managers create IT by integrating multiple data sources, standardizing information flows, and ensuring that continuously generated project information is accessible across organizational boundaries. Through effective resource structuring, fragmented project data are transformed into reliable operational knowledge that reduces uncertainty and improves project visibility. IT therefore represents an important organizational capability through which DTT generates operational value.

The value of IT extends beyond simple information availability. In construction projects, continuously updated operational information enables project managers, contractors, subcontractors, consultants, and clients to develop a shared understanding of project status,

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rapidly identify emerging operational issues, and coordinate timely corrective actions (Anumba *et al.*, 2008). Improved transparency reduces information asymmetry, minimizes misunderstandings between project participants, enhances accountability, and supports faster responses to schedule deviations, quality issues, resource shortages, and safety incidents. Consequently, DTT contributes to operational effectiveness by transforming fragmented information into continuously shared situational awareness across the project network.

Compared with conventional BIM-enabled information management, DTT provides significantly greater operational transparency because project information evolves continuously rather than periodically (Brandín and Abrishami, 2025). This capability is particularly valuable within temporary project organizations where numerous independent stakeholders must coordinate decisions under rapidly changing construction conditions. As project complexity increases, the ability to access reliable, real-time operational information becomes an increasingly important determinant of project performance and operational efficiency.

This distinction is important for IT because DTT extends the periodically updated information environment associated with BIM into a continuously synchronized operational information environment, thereby providing stakeholders with more timely visibility into actual site conditions. Accordingly, organizations adopting DTT are expected to achieve significantly higher levels of IT within construction projects. Thus, we hypothesize that:

H2. DTT has a positive effect on IT.

3.3 Digital twin technology and collaborative decision-making

CDM is essential in construction projects, where multiple stakeholders must coordinate actions under complex and dynamic conditions. However, decision-making is often constrained by fragmented information, a lack of real-time data, and limited coordination among project participants (Allaoui *et al.*, 2019). DTT provides an integrated digital environment that supports shared understanding and joint decision processes by enabling real-time data access, visualization, and simulation of project activities. This allows stakeholders to evaluate different scenarios, anticipate potential issues, and make coordinated decisions more effectively (Haesevoets *et al.*, 2021).

Construction projects differ from many other organizational settings because critical operational decisions are distributed across numerous independent organizations, including owners, contractors, subcontractors, consultants, suppliers, and project managers. These stakeholders often possess different objectives, information systems, and decision responsibilities while simultaneously operating within rapidly changing physical environments. As a result, effective project execution depends not only on information availability but also on the ability of diverse project participants to interpret evolving site conditions collectively and coordinate timely operational responses.

Although BIM provides an important platform for multidisciplinary coordination during planning and design, CDM during construction execution requires continuously updated operational information that reflects current site conditions. Because BIM models are generally revised periodically, they may not always capture unexpected changes occurring during project execution. In contrast, DTT continuously synchronizes physical construction activities with digital representations through sensors, IoT devices, drones, wearable technologies, and equipment monitoring systems (Oyediran *et al.*, 2025). This continuous synchronization enables all project participants to evaluate the same operational information simultaneously, thereby supporting collective situational awareness and reducing inconsistencies in project decision-making.

Recent studies highlight that DTT significantly enhances CDM by improving information sharing, coordination, and interaction among stakeholders (Görsch *et al.*, 2026). DTT

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platforms enable more accurate and efficient decision-making through real-time data integration and visualization in construction environments (Liu *et al.*, 2025). DTT applications have been shown to support collaboration and information exchange among project participants, leading to better coordinated operational decisions (Wang *et al.*, 2026). DTT-based systems facilitate CDM by enabling multiple actors to interact with shared digital models and jointly evaluate operational alternatives (Arowoiya *et al.*, 2026). These capabilities strengthen coordination and improve the quality of decisions in complex project environments.

The operational value of DTT extends beyond information sharing by supporting continuous collaboration throughout project execution. For example, when unexpected schedule delays, equipment failures, material shortages, or safety incidents occur, continuously synchronized digital twins allow contractors, subcontractors, engineers, and project managers to visualize current site conditions, evaluate alternative response strategies, assess their likely consequences through simulation, and collectively determine appropriate corrective actions. Consequently, DTT shortens decision cycles, reduces uncertainty, improves coordination across organizational boundaries, and enhances the consistency of operational decisions.

From the perspective of ROT, CDM represents the resource bundling and leveraging stages of resource orchestration. After managers structure digital resources by establishing integrated information infrastructures, they must combine continuously updated operational information with the expertise, experience, and coordination routines of multiple project participants. Through this process, digital information becomes a shared organizational capability that enables coordinated decision-making rather than remaining an isolated technological resource. Thus, DTT creates value not simply by providing information but by enabling managers to orchestrate knowledge, expertise, and collaborative processes across temporary project organizations.

The collaborative capabilities enabled by DTT are particularly valuable in construction because project activities are highly interdependent. Decisions made by one stakeholder frequently influence the work of numerous other participants. Continuous access to synchronized operational information enables stakeholders to anticipate downstream impacts, coordinate resource allocation, align work schedules, and resolve conflicts before they escalate into costly project disruptions. Compared with conventional digital platforms, DTT therefore supports a more proactive and integrated approach to collaborative project management by facilitating real-time coordination rather than periodic communication.

Similarly, the distinction between BIM and DTT is relevant to CDM because collaborative construction decisions depend on current operational conditions; DTT's continuous synchronization therefore provides a more dynamic basis for collective decision-making than periodically updated project information. Accordingly, DTT enhances CDM by providing continuously updated operational intelligence that enables multiple project participants to jointly evaluate evolving construction conditions and coordinate timely responses across organizational boundaries.

H3. DTT has a positive effect on CDM.

3.4 The mediating role of IT and CDM

While DTT improves operational processes, its impact on OME is not only direct but also occurs through important organizational mechanisms (Warner and Wäger, 2023). This argument is consistent with ROT, which suggests that technological resources create value only when they are effectively structured, bundled, and leveraged through complementary organizational processes. In construction projects, the value of digital technologies is realized when they enhance how information is shared and how decisions are made among stakeholders (Patil *et al.*, 2024). IT and CDM play key roles in translating digital capabilities into operational outcomes.

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ROT argues that organizations rarely obtain performance benefits from technological investments through direct effects alone. Instead, managers create value by transforming technological resources into organizational capabilities that improve operational processes. Accordingly, DTT should not be viewed as an isolated technological resource but as an enabling digital platform whose operational benefits depend on the organizational mechanisms through which continuously generated information is interpreted, shared, and applied throughout construction projects. This perspective provides the theoretical rationale for examining IT and CDM as mediating mechanisms linking DTT to OME.

DTT enables immediate data integration, visualization, and continuous monitoring of project activities, which significantly improves IT (Bi *et al.*, 2024). When stakeholders have access to accurate and up-to-date information, they can better understand project conditions, reduce uncertainty, and coordinate their actions more effectively. This improved transparency facilitates smoother communication and reduces errors, which ultimately enhances operational efficiency (Jia *et al.*, 2025).

Within construction environments, improved IT is particularly important because operational decisions frequently depend on rapidly changing site conditions. Continuous synchronization between physical construction activities and their digital representations enables project participants to identify deviations from planned schedules, monitor equipment utilization, verify material availability, and detect emerging safety risks in real time. Such transparency reduces information asymmetry among contractors, subcontractors, consultants, and project managers, allowing decisions to be based on consistent and reliable operational information rather than fragmented reports. Consequently, IT represents the primary mechanism through which DTT transforms real-time digital resources into coordinated operational activities.

At the same time, DTT platforms support CDM by providing a shared digital environment where stakeholders can jointly evaluate project scenarios, solve problems, and align their decisions. This collaborative process improves the quality and speed of decision-making, leading to more efficient execution of project activities (Cheng *et al.*, 2016).

CDM complements IT because access to high-quality information alone does not automatically improve project performance. Construction projects involve multiple organizations with different responsibilities, objectives, and expertise. Therefore, project participants must collectively interpret operational information, evaluate alternative courses of action, negotiate priorities, and coordinate implementation. DTT facilitates this process by providing continuously updated operational intelligence and predictive simulations that enable stakeholders to assess the consequences of alternative decisions before implementing corrective actions. As a result, CDM transforms transparent information into coordinated operational responses that improve project execution.

The complementary roles of IT and CDM closely reflect the sequential resource orchestration process proposed by ROT. IT represents the outcome of resource structuring because DTT-generated information must first be integrated, standardized, and made accessible across project participants. CDM subsequently reflects resource bundling and leveraging because transparent information is combined with stakeholder expertise, managerial experience, and coordination routines to support collective operational decisions. Thus, DTT is not expected to improve OME solely through its technological functionality. Rather, its operational value is expected to emerge through the transformation of continuously generated digital information into transparent information environments and coordinated decision processes. Through this sequence, technological resources are progressively transformed into organizational capabilities that improve operational efficiency.

Recent studies support this mediating mechanism. DTT systems enhance CDM and coordination by providing shared, present information environments that improve collaboration and operational outcomes (Xue *et al.*, 2025; Yin *et al.*, 2024). Similarly, research shows that DTT improves information exchange and stakeholder coordination, which in turn enhances project performance (Singh *et al.*, 2023).

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Compared with conventional BIM-enabled information management, DTT provides greater potential for these mediating mechanisms because operational information is continuously synchronized with physical construction activities rather than periodically updated. This continuous synchronization strengthens both IT and CDM by ensuring that all project participants base operational decisions on the same current project conditions. Accordingly, IT and CDM represent complementary organizational mechanisms through which the technological capabilities of DTT can be translated into operational efficiency. IT and CDM are not treated merely as intermediate technological or organizational outcomes of DTT. They represent the mechanisms through which DTT-enabled information is translated into operational improvements. Real-time IT provides project participants with timely and reliable visibility into construction conditions, while CDM enables stakeholders to collectively interpret this information and coordinate operational responses. These mechanisms are therefore expected to connect DTT capabilities with improvements in OME.

OME is therefore expected to be supported through improved transparency and CDM. Accordingly, IT and CDM are expected to mediate the relationship between DTT and OME. Thus, we hypothesize that:

H4. IT positively influences OME.

H5. CDM positively influences OME.

Given H2–H5, we further examine whether IT and CDM mediate the relationship between DTT and OME through bootstrapped indirect effects.

3.5 The moderating role of organizational digital readiness

The relationship between DTT and OME is not uniform across organizations because the effectiveness of digital technologies depends on the extent to which firms are prepared to adopt and utilize them (Zheng, 2024). While DTT enables real-time monitoring, data integration, and improved coordination of construction activities, these capabilities do not automatically translate into operational efficiency. The ability to execute processes efficiently while optimizing time, cost, and resource utilization depends on the organizational context in which DTT is implemented. ODR plays a critical role by strengthening the impact of DTT on operational outcomes (Kumar and Sharma, 2025). Organizations with higher levels of digital readiness possess the necessary infrastructure, technical skills, and managerial support to integrate DTT systems effectively into their workflows, enabling better use of real-time data, improved coordination, and more efficient decision-making (Machado *et al.*, 2020). This argument is consistent with absorptive-capacity research, which suggests that organizations with stronger knowledge-processing capabilities are better positioned to capture value from technological innovations (Cohen and Levinthal, 1990; Zahra and George, 2002). In contrast, organizations with low digital readiness may struggle to implement and utilize digital technologies effectively, thereby limiting the potential benefits of DTT adoption (Tenggono *et al.*, 2025). For example, insufficient digital infrastructure may prevent real-time data integration, inadequate employee skills may reduce effective system utilization, and weak managerial support may hinder organizational adoption and cross-functional collaboration. As a result, the information and coordination advantages offered by DTT may not translate into meaningful operational improvements.

At the same time, the relationship between digital readiness and the marginal benefits of additional digital technologies may not necessarily be linear. Research on digital transformation paradoxes suggests that increasing levels of digitalization can create both enabling and constraining effects, including greater technological complexity, coordination demands, and tensions between established and newly introduced digital practices. Similarly, organizations with highly developed digital infrastructures may already possess information systems, analytics capabilities, and technology-enabled processes that partially overlap with some capabilities provided by DTT. Such resource redundancy may reduce the incremental

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efficiency gains obtained from introducing additional digital technology. Organizational inertia may further constrain the realization of these gains when highly digitally mature organizations have established routines, technological architectures, and coordination practices that are difficult or costly to modify. Thus, although high ODR can facilitate the implementation and utilization of DTT, it may also create conditions under which the marginal contribution of DTT becomes less pronounced.

These potential dampening effects do not negate the enabling role of ODR. Rather, they represent a competing theoretical possibility concerning the extent to which existing digital maturity changes the incremental value of DTT. The present study therefore retains the positive moderation argument because, from a resource orchestration perspective, organizations with stronger digital readiness are expected to possess greater capabilities to structure, bundle, and leverage DTT-enabled resources. However, the possibility that high baseline digital maturity may reduce the marginal gains from DTT is acknowledged as an alternative explanation for the boundary conditions surrounding this relationship.

From the perspective of ROT, organizational resources create value only when managers possess the capabilities required to structure, bundle, and leverage them effectively. Accordingly, DTT should not be expected to improve OME uniformly across construction organizations because firms differ substantially in their ability to deploy digital technologies, integrate real-time information, and coordinate technology-enabled work processes. ODR therefore represents an enabling organizational capability that determines how effectively digital resources are transformed into operational outcomes.

Construction projects present particularly demanding implementation environments because they involve temporary project organizations, geographically dispersed worksites, multiple independent stakeholders, heterogeneous digital systems, and continuously changing operational conditions. Under these circumstances, the successful implementation of DTT requires considerably more than technological investment alone. Organizations must establish reliable digital infrastructure, develop employee competencies, standardize digital workflows, promote collaboration across organizational boundaries, and cultivate managerial commitment to digital transformation. Without these complementary organizational capabilities, the real-time and bidirectional capabilities of DTT cannot be fully exploited during project execution.

This moderating relationship also complements the mediating mechanisms proposed earlier. While IT and CDM explain how DTT generates operational value, ODR explains when and under what organizational conditions these value-creation processes are likely to be strongest. Construction organizations possessing greater digital readiness are therefore expected to derive greater operational benefits from DTT because they are better able to convert continuously synchronized digital information into coordinated managerial actions.

The proposed moderation is consistent with both ROT and absorptive-capacity theory, which argue that organizations possessing stronger complementary capabilities derive greater value from technological resources than organizations lacking the necessary organizational foundations. Accordingly, ODR strengthens the positive influence of DTT on OME by improving the effectiveness with which digital resources are orchestrated within project-based construction environments.

Thus, we hypothesize that:

H6. ODR positively moderates the relationship between DTT and OME.

Although ODR may theoretically influence multiple stages of resource orchestration, the present study focuses on its moderating effect on the DTT–OME relationship because this relationship captures the overall conversion of digital twin capabilities into operational performance. This parsimonious specification is consistent with ROT by treating digital readiness as a higher-order organizational capability that enhances the effectiveness of the entire resource orchestration process rather than a capability operating at individual intermediary stages.

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4. Methodology design

4.1 Sample and data collection

The selection and verification of measurement variables are critical tasks for data collection when performing empirical analysis. Therefore, the choice of suitable variables plays a crucial role in ensuring the consistency of results and overall validity. In this study, data were collected via a web-based survey from construction professionals whose firms operate in Shandong Province, China. Shandong was selected because it is one of the most economically developed provinces in China, with strong growth in infrastructure and construction activities. The region has also increasingly adopted digital technologies, such as BIM and related systems, to improve construction project performance, making it a relevant context for this study (Shandong Provincial Bureau of Statistics, 2023). The survey was conducted using an online platform **(**https://www.wjx.cn/**)**, which is widely used for academic and industry research in China. Before distributing the questionnaire, it was carefully pre-tested with five academic experts and five industry professionals to ensure the clarity and relevance of the items.

A purposive sampling approach was adopted as the study required respondents with direct knowledge of construction operations and experience with digital technologies. To increase access to qualified respondents, the survey link was further disseminated through professional networks and industry-specific WeChat groups using a controlled snowball sampling technique. This approach is consistent with prior studies examining emerging digital technologies in specialized professional contexts where comprehensive sampling frames are unavailable.

The respondents were screened based on specific criteria for eligibility. Participants needed to be working in a decision-making or technical role (e.g. project manager, construction manager, site engineer, or BIM specialist) within construction projects and have experience with digital technologies. The questionnaire was distributed from January 4 to March 30, 2026, through professional networks, email, and WeChat groups commonly used by construction professionals. A total of 450 questionnaires were distributed, of which 380 were returned, resulting in a response rate of 84.4%. After excluding 67 questionnaires due to incomplete, duplicate, or inconsistent responses, 313 valid responses remained for analysis.

An *a priori* power analysis using G*Power 3.1 (Faul *et al.*, 2009), assuming a medium effect size ($f^2 = 0.15$), a significance level of $\alpha = 0.05$, statistical power of 0.95, and five predictors, indicated a minimum sample size of approximately 138 observations. As the final sample comprised 313 valid responses, the study exceeded the recommended sample size by a substantial margin, providing adequate statistical power for detecting the hypothesized relationships.

The questionnaire was accompanied by an informed consent statement explaining the study purpose, voluntary nature of participation, anonymity, and confidentiality of responses. Respondents were informed that they could withdraw at any time without penalty. The study received ethical approval from the Institutional Review Board (IRB) of the authors' institution (IRB Approval No. 25-019; approved on 29 October 2024) and was conducted in accordance with the approved ethical protocol for research involving human participants. The study complied with institutional ethical research guidelines for human-subject research. To assess the possibility of non-response bias, early and late respondents were compared on key demographic characteristics and construct means. No statistically significant differences were observed, suggesting that non-response bias was unlikely to affect the findings.

4.2 Measurement and validation

The measurement items used in this study were adapted from established literature to ensure reliability and validity. All constructs were measured using a 5-point Likert scale, ranging from (1) "strongly disagree" to (5) "strongly agree". The questionnaire was initially developed in English and then translated into Chinese to ensure clarity for respondents. To ensure linguistic equivalence, a back-translation procedure was employed. The original English

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questionnaire was translated into Chinese by a bilingual researcher and subsequently translated back into English by an independent bilingual expert. Any discrepancies between the original and back-translated versions were reviewed and resolved through discussion, thereby improving translation accuracy and content equivalence. To ensure content validity, the questionnaire was reviewed by academic experts and industry professionals, and minor modifications were made based on their feedback. Table 2 presents the constructs and measurement items and corresponding sources.

Although respondent position, industry segment, work experience, and other demographic characteristics were collected, these variables were not incorporated as formal control variables in the structural model. The primary objective of this study was theory testing and prediction of the hypothesized relationships rather than estimating the net effects after accounting for organizational heterogeneity. Consequently, the structural model was estimated without control variables to preserve model parsimony. Nevertheless, the absence of control variables represents a limitation of the present study and should be considered when interpreting the findings. Future research should explicitly incorporate organizational and project-related controls, such as firm size, project complexity, project type, respondent tenure, and prior digital twin experience, to further examine the robustness of the reported relationships.

Although several OME items refer to the operational outcomes associated with DTT implementation, the construct was conceptualized and measured as operational efficiency rather than technology adoption itself. Specifically, the DTT construct captures the organization's capability to utilize digital twin functionalities (e.g. visibility, coordination, real-time monitoring, flexibility, and decision support), whereas OME reflects perceived operational outcomes such as process improvement, cost reduction, quality enhancement, and productivity gains.

Nevertheless, because several OME indicators explicitly reference digital technologies, some degree of conceptual proximity between the predictor (DTT) and outcome (OME) cannot be entirely excluded. Such proximity may contribute to stronger observed relationships than would be expected if operational efficiency were measured using technology-neutral indicators. Accordingly, the findings should be interpreted with appropriate caution, particularly regarding the magnitude of the estimated effects.

The explicit reference to DTT in selected OME items reflects the study's objective of evaluating efficiency outcomes attributable to digital twin deployment rather than measuring general operational efficiency in isolation. Consistent with prior technology-performance research, outcome measures may reference the focal technology when assessing perceived operational improvements resulting from its implementation.

To evaluate whether this conceptual proximity resulted in substantial empirical overlap, multiple diagnostic procedures were conducted, including convergent validity, discriminant validity (Fornell–Larcker criterion, HTMT, and cross-loadings), and collinearity diagnostics. Overall, these assessments indicated that DTT and OME remained empirically distinguishable constructs. However, these statistical diagnostics cannot eliminate the possibility of measurement contamination arising from the wording of several OME indicators. Consequently, future studies are encouraged to employ more outcome-pure measures of operational efficiency, including objective project-performance indicators (e.g. cost overruns, schedule adherence, productivity, quality defects, and rework), to further validate the relationships reported in this study.

4.3 Statistical techniques

This study employed Partial Least Squares Structural Equation Modeling (PLS-SEM) using SmartPLS, along with Necessary Condition Analysis (NCA), to test the proposed model. PLS-SEM is suitable for analyzing complex relationships, including mediation and moderation effects, and focuses on examining the net impact of i*ndependent variables* on the *dependent*

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Table 2. Constructs and indicators

Constructs	Items	Indicator	Adapted from
Digital twin technology	*DTT1*	Visibility	Singh *et al.* (2023), Su *et al.* (2023)
	DTT2	Coordination	
	DTT3	Real-time monitoring	
	DTT4	Transparency	
	DTT5	Flexibility	
	DTT6	Risk awareness	
	DTT7	Velocity	
	DTT8	Better decision-making	
	DYT9	Better demand forecasting	
Information transparency	*IT1*	The system tracks project activities and progress in real time	Al-Jabri and Roztocki (2015), Long *et al.* (2024)
	IT2	The system provides clear information on project rules and regulations	
	IT3	The system offers transparency on decisions and actions taken by project managers	
	IT4	The system monitors financial expenditures, ensuring cost transparency	
	IT5	The system shares real-time performance data for assessing progress	
	IT6	The system promotes openness in processes like resource allocation and scheduling	
	IT7	The system enhances transparency by improving communication among stakeholders	
Collaborative decision-making	*CDM1*	Our company collaboratively solves construction-related challenges with project stakeholders	Lai *et al.* (2020), Devi *et al.* (2023)
	CDM2	Our company works together with stakeholders to coordinate resource and equipment needs	
	CDM3	Our company jointly plans and coordinates project schedules and timelines with project partners	
	CDM4	Our company collaborates with stakeholders to analyze market trends and project requirements	
	CDM5	Our company cooperates with project partners to develop innovative solutions for construction projects	
Organizational digital readiness	*ODR1*	We understand that adopting new digital technologies can improve project outcomes and operational performance	Zhen *et al.* (2021), Zhang *et al.* (2026)
	ODR2	Most staff members are willing to embrace new digital tools, and it is easy to modify existing procedures to meet new technological demands	
	ODR3	Staff members frequently ask questions and express concerns when new digital technologies are introduced into the organization	
	ODR4	When new technologies are necessary, management provides a clear plan for integrating these technologies into the organization's workflow	
	ODR5	Staff are encouraged to discuss and explore innovative digital tools and practices to improve operational efficiency	
	ODR6	Staff members adapt quickly when they have to shift focus to accommodate technological changes in the organization	

(*continued*)

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Table 2. Continued

Constructs	Items	Indicator	Adapted from
Operations management efficiency	*OME1*	We leverage DTT to improve project management processes	Susilawati *et al.* (2023), Nasir and Wen (2026)
	OME2	We continually enhance employee skills to use digital tools effectively	
	OME3	We promote efficient project delivery through digital tools	
	OME4	Adopting DTT has reduced operational costs in our projects	
	OME5	DTT has improved the quality of project operations	
	OME6	Adopting DTT has freed up time for staff to focus on project outcomes	

Source(s): Authors' own creation/work

variable (Hair *et al.*, 2017a). The analysis was conducted in two stages: assessment of the *measurement model* (reliability and validity) and evaluation of the *structural model* (hypothesis testing). PLS-SEM was selected because the study aims to predict OME and examine multiple mediating and moderating relationships simultaneously. Furthermore, PLS-SEM is particularly appropriate for exploratory and theory-extension research involving emerging technologies such as DTT.

In contrast, NCA focuses on identifying the necessary conditions required for achieving the desired outcome. In this study, the desired outcome is a high level of OME. NCA evaluates whether certain organizational conditions must reach a minimum threshold before higher levels of OME can be achieved. PLS-SEM, which assesses whether variables contribute to OME on average, NCA determines whether insufficient levels of a particular condition would prevent organizations from attaining higher levels of OME. The use of NCA complements ROT by identifying whether specific organizational capabilities represent indispensable prerequisites for achieving operational efficiency, rather than merely contributing to efficiency improvements.

The structural model was estimated without the inclusion of control variables. Accordingly, the reported path coefficients represent the direct relationships among the focal theoretical constructs and should not be interpreted as adjusted estimates that account for potential organizational or project-level heterogeneity. This modelling decision is acknowledged as a limitation and provides an opportunity for future research to evaluate the stability of the proposed relationships across different organizational contexts.

5. Results and analysis

5.1 Empirical results

The results of this study are based on a sample of 313 respondents whose characteristics are presented in Table 3. Specifically, the sample is composed of 63.3% males and 36.7% females. In terms of age, 29.4% of the respondents are between 18 and 24 years old, 45.0% are between 25 and 34 years old, 18.5% are between 35 and 44 years old, and 7.0% are above 45 years old. This indicates that 74.4% of the respondents are up to 34 years old, while 92.9% are up to 44 years old. In terms of qualifications, 65.2% hold an undergraduate degree, 24.9% hold a graduate degree, and 9.9% possess a postgraduate qualification.

Regarding professional designation, 41.9% of the respondents are project managers, 26.8% are construction managers, 16.6% are BIM engineers, and 14.6% are IT specialists. In terms of industry affiliation, responses were collected from professionals working in building

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Table 3. Demographic profile of the respondents

Respondent profile		*(n = 313)* *Frequency*	*(%)*
Gender	Male	198	63.3
	Female	115	36.7
	Total	*313*	*100%*
Age	18–24	92	29.4
	25–34	141	45.0
	35–44	58	18.5
	Above 45	22	7.0
	Total	*313*	*100%*
Qualifications	Undergraduate	204	65.2
	Graduate	78	24.9
	Postgraduate	31	9.9
	Total	*313*	*100%*
Designation	Project manager	131	41.9
	Construction manager	84	26.8
	BIM engineer	52	16.6
	IT specialist	46	14.7
	Total	*313*	*100%*
Industry	Building construction	132	42.2
	Infrastructure projects	79	25.2
	Construction consulting	44	14.1
	Engineering/contracting firms	58	18.5
Work experience (in years)	*Total*	*313*	*100%*
	5 Years	78	24.9
	6–10 Years	163	52.0
	11–15 Years	72	23.0
	Total	*313*	*100%*

Source(s): Authors' own creation/work

construction (42.2%), infrastructure projects (25.2%), construction consulting (14.1%), and engineering/contracting firms (18.5%). With respect to work experience, 24.9% of the respondents have 5 years of experience, 52.0% have 6–10 years of experience, and 23.0% have 11–15 years of experience, showing that the majority of respondents possess considerable industry experience.

The respondent profile indicates that the sample consisted primarily of experienced construction professionals occupying managerial and technical positions. This enhances confidence that respondents possessed sufficient knowledge of construction operations and digital technologies to provide informed assessments of the study constructs.

5.2 Measurement model (outer model)

5.2.1 Convergent validity analysis. In the preliminary phase, the reliability and validity (both convergent and discriminant) of the measurement model were assessed following the recommendations of Hair *et al.* (2019). Table 5 reports the indicator loadings together with Cronbach's alpha (α), rho_a, composite reliability (rho_c), average variance extracted (AVE), and variance inflation factor (VIF). Cross-loadings, presented in Table 4, compare each indicator's loading on its assigned construct with its loadings on all other constructs and provide an additional assessment of discriminant validity.

The internal consistency of the constructs was evaluated using Cronbach's alpha, rho_a, and rho_c. As shown in Table 5, all three reliability coefficients exceeded the recommended threshold of 0.70, while all AVE values were greater than 0.50 (Hair *et al.*, 2017b), confirming satisfactory internal consistency reliability and convergent validity.

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Table 4. Cross-loading

Construct	CDM	DTT	IT	ODR	OME
CDM1	*0.716*	0.364	0.490	0.306	0.554
CDM2	*0.733*	0.496	0.322	0.509	0.492
CDM3	*0.742*	0.463	0.581	0.260	0.508
CDM4	*0.810*	0.507	0.456	0.402	0.225
CDM5	*0.795*	0.336	0.395	0.447	0.491
DTT1	0.699	*0.752*	0.518	0.386	0.279
DTT2	0.612	*0.726*	0.591	0.435	0.203
DTT3	0.300	*0.747*	0.296	0.457	0.399
DTT4	0.427	*0.778*	0.520	0.303	0.639
DTT5	0.416	*0.679*	0.413	0.265	0.426
DTT6	0.390	*0.672*	0.400	0.264	0.446
DTT7	0.431	*0.741*	0.518	0.328	0.521
DTT8	0.509	*0.721*	0.511	0.229	0.443
DTT9	0.587	*0.810*	0.582	0.333	0.445
IT1	0.711	0.562	*0.688*	0.305	0.554
IT2	0.434	0.496	*0.722*	0.508	0.493
IT3	0.338	0.461	*0.678*	0.259	0.504
IT4	0.310	0.504	*0.759*	0.400	0.621
IT5	0.590	0.329	*0.793*	0.442	0.483
IT6	0.634	0.301	*0.786*	0.453	0.422
IT7	0.204	0.549	*0.825*	0.514	0.552
ODR1	0.220	0.241	0.253	*0.674*	0.275
ODR2	0.323	0.476	0.407	*0.698*	0.285
ODR3	0.234	0.278	0.290	*0.779*	0.328
ODR4	0.241	0.221	0.264	*0.728*	0.291
ODR5	0.497	0.280	0.223	*0.619*	0.213
ODR6	0.249	0.267	0.270	*0.750*	0.282
OME1	0.400	0.347	0.396	0.457	*0.799*
OME2	0.427	0.578	0.520	0.303	*0.639*
OME3	0.423	0.476	0.407	0.398	*0.785*
OME4	0.643	0.513	0.387	0.464	*0.837*
OME5	0.630	0.572	0.567	0.319	*0.653*
OME6	0.424	0.477	0.410	0.497	*0.788*

Note(s): Collaborative Decision-Making, Digital Twin Technology, Information Transparency, Organizational Digital Readiness, Operations Management Efficiency

Source(s): Authors' own creation/work

Nevertheless, several indicators exhibited localized cross-loading concerns, including DTT5, DTT6, IT1, IT3, ODR1, ODR2, ODR5, OME2, and OME5. Among these, IT1 deserves particular attention because it loaded slightly higher on CDM (0.711) than on its intended construct, IT (0.688). This pattern suggests that respondents may perceive transparent information sharing and CDM as highly intertwined within digitally enabled construction projects. Although this item does not satisfy the strict cross-loading preference, its loading on the intended construct remained acceptable (>0.60), the overall construct demonstrated satisfactory internal consistency and convergent validity, and both the Fornell-Larcker and HTMT criteria supported discriminant validity at the construct level. Consequently, the item was retained to preserve the theoretical coverage of the IT construct. Nevertheless, this localized overlap indicates that future studies should refine or reword this indicator to achieve clearer conceptual separation between transparency and CDM. Similar considerations apply to the remaining indicators exhibiting moderate cross-loadings. Accordingly, these results indicate localized measurement overlap rather than pervasive discriminant validity failure, but they warrant cautious interpretation of the structural relationships.

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Table 5. Measures of internal consistency, reliability, and convergent validity

Constructs	Items	Loading	Cronbach's alpha (α)	rho_a	Composite reliability (rho_c)	AVE	VIF
Collaborative decision-making	CDM1	0.716	0.818	0.827	0.872	0.578	1.481
	CDM2	0.733					1.575
	CDM3	0.742					2.196
	CDM4	0.810					2.476
	CDM5	0.795					1.608
Digital twin technology	DTT1	0.752	0.897	0.908	0.914	0.544	2.049
	DTT2	0.726					2.002
	DTT3	0.747					3.770
	DTT4	0.778					2.543
	DTT5	0.679					2.055
	DTT6	0.672					2.156
	DTT7	0.741					2.452
	DTT8	0.721					2.477
	DTT9	0.810					3.114
Information transparency	IT1	0.688	0.872	0.881	0.901	0.566	1.613
	IT2	0.722					1.816
	IT3	0.678					2.178
	IT4	0.759					2.501
	IT5	0.793					1.967
	IT6	0.786					2.079
	IT7	0.825					2.454
Organizational digital readiness	ODR1	0.674	0.837	0.969	0.858	0.504	1.871
	ODR2	0.698					1.088
	ODR3	0.779					3.896
	ODR4	0.728					2.152
	ODR5	0.619					1.771
	ODR6	0.750					3.793
Operations management efficiency	OME1	0.799	0.845	0.854	0.887	0.568	1.958
	OME2	0.639					2.128
	OME3	0.785					3.887
	OME4	0.837					3.223
	OME5	0.653					1.508
	OME6	0.788					2.847

Source(s): Authors' own creation/work

As shown in Table 5, the Cronbach's alpha, rho_a, rho_c, and AVE values exceeded the recommended thresholds for all constructs. For the ODR construct, the value reported in the rho_a column was 0.969, whereas the corresponding value in the composite reliability column was 0.858. Although rho_a exceeded composite reliability for this construct, all reliability indices remained above the recommended threshold of 0.70, and the AVE value (0.504) exceeded the recommended threshold of 0.50, supporting the acceptable reliability and convergent validity of the ODR construct. Nevertheless, the relatively large difference between rho_a and composite reliability suggests that the measurement properties of this construct should be further examined in future validation studies.

5.2.2 Discriminant validity analysis. Discriminant validity is a critical assessment in PLS-SEM, as it establishes the originality and distinctiveness of each construct (Aziz and Qazi, 2026). In this study, discriminant validity was evaluated using three approaches: the Fornell–Larcker criterion, the Heterotrait–Monotrait Ratio (HTMT), and the cross-loadings criterion. According to the Fornell–Larcker criterion, the square root of the AVE for each construct exceeded its correlations with all other constructs, thereby confirming discriminant validity

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(Fornell and Larcker, 1981). Furthermore, the HTMT values were below the recommended thresholds of 0.85 and 0.90, providing additional evidence of adequate discriminant validity (Hair *et al*., 2017b). The VIF values were also below the recommended cutoff value of 3.3, indicating the absence of common method bias (CMB) in the model. The detailed results are presented in Tables 6 and 7.

To address concerns regarding CMB, the privacy and confidentiality of respondents were ensured (Podsakoff and Organ, 1986). Item ambiguity was minimized by using well-established measurement scales, and the questionnaire was pre-tested for clarity. CMB may occur when both dependent and independent variables are collected from the same respondents using a single survey method (Jordan and Troth, 2020). Therefore, the full collinearity VIF approach was applied to assess the presence of CMB, and all values were found to be within the acceptable threshold. Furthermore, Harman's single-factor test was conducted (Fuller *et al*., 2016). The results showed that the first factor accounted for 45.5% of the variance, which is lower than the 50% threshold.

Although the Fornell-Larcker criterion, HTMT ratios, reliability statistics, and collinearity diagnostics all supported acceptable measurement quality, several localized cross-loading patterns remained evident. In particular, IT1 loaded marginally higher on CDM than on IT, suggesting that respondents did not clearly distinguish IT from CDM in this context. Rather than indicating a fundamental measurement problem, this finding likely reflects the close conceptual relationship between transparent information exchange and collaborative work practices in construction projects. Nevertheless, this overlap remains a measurement

Table 6. Fornell-Larcker criterion (Discriminant validity using AVE)

Constructs	CDM	DTT	IT	ODR	OME
CDM	*0.760*				
DTT	0.570	*0.737*			
IT	0.604	0.410	*0.752*		
ODR	0.474	0.559	0.596	*0.710*	
OME	0.379	0.331	0.433	0.654	*0.754*

Note(s): Collaborative Decision-Making, Digital Twin Technology, Information Transparency, Organizational Digital Readiness, Operations Management Efficiency *(Square roots of AVEs shown on diagonal)*

Source(s): Authors' own creation/work

Table 7. Discriminant validity using heterotrait-monotrait ratio (HTMT)

Constructs	CDM	DTT	IT	ODR	OME	ODR x DTT
CDM						
DTT	0.438					
IT	0.549	0.662				
ODR	0.491	0.410	0.516			
OME	0.231	0.422	0.366	0.580		
ODR x DTT	0.332	0.512	0.433	0.365	0.568	

Note(s): Collaborative Decision-Making, Digital Twin Technology, Information Transparency, Organizational Digital Readiness, Operations Management Efficiency; ODR × DTT represents the latent interaction construct between ODR and DTT created for moderation analysis. HTMT values are reported between the interaction construct and each latent construct to assess discriminant validity. All HTMT values are below the recommended threshold of 0.90, indicating satisfactory discriminant validity (Kock, 2017)

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limitation, and future studies should refine the wording of IT1 and related indicators to improve construct distinctiveness. Accordingly, the measurement model was considered adequate for structural analysis, but the reported relationships should be interpreted with appropriate caution because some conceptual overlap cannot be completely excluded.

*5.3 Structural model results (*inner model)

In the structural model, exogenous and endogenous variables are linked through hypothesized causal relationships (Hair *et al.*, 2017a). The structural model assessment included collinearity assessment, explanatory power (R^2), effect size (f^2) and β-values (path coefficient) and hypothesis testing through bootstrapping procedures.

5.3.1 Collinearity analysis. Increased correlations among predictor constructs are generally undesirable; therefore, collinearity assessment was conducted before evaluating the structural model. All VIF values were below the recommended threshold of 3.3, indicating the absence of multicollinearity concerns. The results suggest that the study constructs contribute independently to the proposed model. The corresponding VIF values are presented in Table 5.

5.3.2 Bootstrap analysis evaluation. A bootstrapping procedure with 5,000 resamples was performed using Smart PLS 4.0 to assess the significance of the proposed hypotheses (Kock, 2015). The significance of the structural relationships was evaluated using path coefficients (β), *t*-values, and *p*-values. The proposed relationships among DTT, CDM, IT, ODR and OME were statistically significant. The detailed results of the hypothesis testing are presented in Table 8 and Figure 2.

Although statistically significant, the magnitude of the IT → OME relationship is considerably larger than is typically reported in organizational research. One plausible explanation is the close conceptual relationship between IT and perceived operational efficiency in digitally enabled construction projects. Furthermore, several OME indicators explicitly refer to operational improvements resulting from digital technologies, which may have strengthened the observed association. Consequently, while the statistical evidence supports a positive relationship, the magnitude of the coefficient should be interpreted cautiously rather than viewed as evidence of an almost deterministic effect.

Similarly, the very high R^2 value observed for OME (0.911) indicates that the model explains a substantial proportion of the variance in perceived operational efficiency. Although high explanatory power is desirable, such values are uncommon in behavioral and organizational survey research. Therefore, the possibility that common respondent perceptions and conceptual proximity among several constructs contributed to this high explanatory power cannot be completely excluded, despite the satisfactory validity and collinearity diagnostics.

Table 8. Hypotheses and path coefficients

Hypothesis	Construct paths	β	*T* value	*p* value	Results
H1	DTT → OME	0.222	6.602	0.000	Supported
H2	DTT → IT	0.810	40.201	0.000	Supported
H3	DTT → CDM	0.770	34.067	0.000	Supported
H4	IT → OME	0.925	10.446	0.000	Supported
H5	CDM → OME	0.263	3.445	0.001	Supported
H6	ODR x DTT → OME	−0.056	3.203	0.002	Not supported in the hypothesized direction

Note(s): Collaborative Decision-Making, Digital Twin Technology, Information Transparency, Organizational Digital Readiness, Operations Management Efficiency

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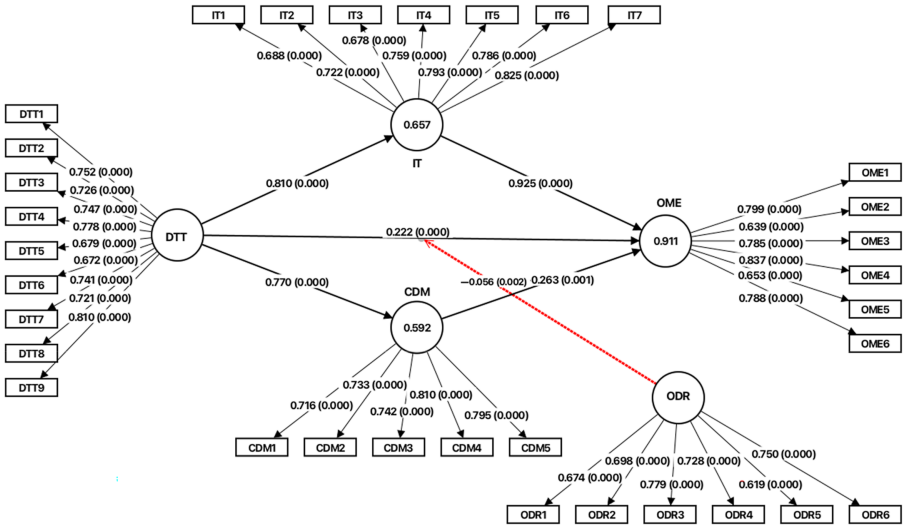


Figure 2. Structural model results with β values. **Source(s):** Authors' own creation/work

The comparison between IT and CDM also provides an important theoretical insight. The considerably stronger mediating role of IT suggests that, within construction environments, timely access to accurate information may represent a more immediate driver of operational efficiency than collaborative interactions alone. Construction projects are characterised by fragmented information flows, temporary organizational arrangements, and multiple stakeholders. Under these conditions, collaboration is likely to be effective only after a shared and transparent information environment has been established.

Although the moderating effect is statistically significant, its direction is opposite to that hypothesized. Accordingly, H6 is not supported in its proposed positive direction. Instead, the findings indicate that ODR significantly moderates the DTT–OME relationship in a negative direction. This result is consistent with the competing boundary-condition explanation introduced in Section 3.5, whereby higher baseline digital readiness may reduce the marginal contribution of DTT because highly digitally prepared organizations may already possess overlapping digital capabilities, established technology-enabled routines, and interconnected systems.

To facilitate interpretation of the interaction effect, the simple slope analysis is presented in Figure 3. The figure illustrates the conditional relationship between DTT and OME at low (−1 SD), mean, and high (+1 SD) levels of ODR. Across all three conditions, the relationship between DTT and OME remains positive, indicating that greater adoption of DTT is consistently associated with higher operational management efficiency. However, the slope of the relationship becomes progressively flatter as ODR increases. Specifically, the steepest positive relationship is observed when ODR is low (−1 SD), whereas the weakest relationship is observed when ODR is high (+1 SD). This pattern is consistent with the negative interaction coefficient ($\beta = -0.056$), demonstrating that higher levels of ODR attenuate the incremental effect of DTT on operational management efficiency rather than amplifying it. Although the moderation effect is relatively small in magnitude, the interaction is statistically significant and suggests diminishing marginal benefits of DTT as organizations become increasingly digitally prepared.

From a theoretical perspective, ROT suggests that digital readiness may influence multiple stages of the resource orchestration process. The present study focuses specifically on the

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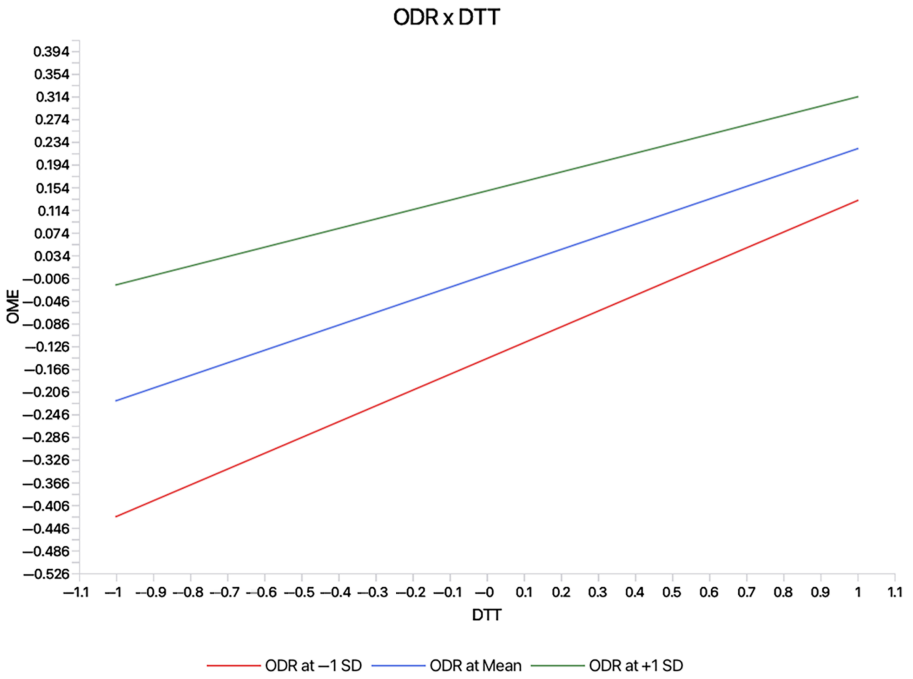


Figure 3. Moderating effect of ODR. **Source(s):** Authors' own creation/work

moderation of the DTT and OME relationship because this path captures the overall conversion of DTT into operational outcomes.

5.3.3 The structural model's exploratory power (R^2*).* The explanatory power of the structural model was assessed by examining the coefficient of determination (R^2) for the endogenous constructs. In addition, effect size (f^2) was evaluated to determine the model's predictive capability and the relative contribution of each exogenous construct. The results of the R^2 and f^2 values are shown in Table 9.

The results indicate substantial explanatory power for CDM ($R^2 = _{0.592}$) and IT ($R^2 = _{0.657}$). OME exhibits a very high R^2 value (0.911). Although this demonstrates excellent predictive capability, it substantially exceeds values typically reported in organizational research using perceptual survey measures. Consequently, the high explanatory power should not be interpreted solely as evidence of superior model performance. Rather, it may also reflect conceptual proximity among several constructs, shared respondent perceptions, and the inclusion of outcome indicators explicitly linked to digital technologies. This should be regarded as residual limitations of the dataset rather than interpreted as definitive evidence of near-deterministic relationships. Future studies should seek to replicate these findings using more technology-neutral operational performance measures, objective project-performance indicators, and independent datasets.

The effect-size analysis indicates exceptionally large effects of DTT on IT ($f^2 = 1.913$) and CDM ($f^2 = 1.452$). These effect sizes are substantially above the conventional benchmarks proposed by Cohen (1988) and Hair *et al.* (2021). This finding suggests that DTT functions as the dominant enabling capability within the proposed theoretical framework. However, the unusually large magnitudes also warrant cautious interpretation because they may partly reflect the conceptual interconnectedness between DTT, IT, and collaborative processes. Consequently, the results should be interpreted as indicating very strong practical relationships rather than precise estimates of effect magnitude.

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Table 9. R^2 and effect size

Constructs	CDM	IT	OME
CDM			0.054
DTT	1.452	1.913	0.189
IT			0.539
ODR			0.154
ODR x DTT			0.037

R^2

	R-square	*R*-square adjusted
CDM	0.592	0.591
IT	0.657	0.656
OME	0.911	0.910

Note(s): Collaborative Decision-Making, Digital Twin Technology, Information Transparency, Organizational Digital Readiness, Operations Management Efficiency

Source(s): Authors' own creation/work

5.4 Mediation test

Following the mediation framework of Nitzl *et al.* (2016) and the updated PLS-SEM guidance of Hair *et al.* (2021), the mediating role of IT and CDM on the relationship between DTT and OME was examined. The mediating roles of IT and CDM, using bootstrapping procedures with 5,000 samples, were assessed regarding indirect effects. The results indicate that both DTT → IT → OME and DTT → CDM → OME indirect paths are statistically significant. These findings suggest that DTT improves OME not only directly but also indirectly through enhanced IT and CDM. As shown in Table 10, IT and CDM significantly mediated the relationship between DTT and OME. The indirect effects of DTT on OME through IT ($\beta = _{0.749}$, $t = _{9.805}$, $p < _{0.001}$) and CDM ($\beta = _{0.202}$, $t = _{3.321}$, $p = _{0.001}$) were both positive and significant. The direct effect of DTT on OME also remained positive and statistically significant ($\beta = 0.222$, $t = 6.602$, $p < 0.001$), while the overall total effect of DTT on OME was positive and significant ($\beta = 0.769$, $t = 35.426$, $p < 0.001$). Furthermore, the Variance Accounted For (VAF) ratio was 97.4% for the indirect effect through IT (0.749/0.769) and 26.3% for the indirect effect through CDM (0.202/0.769). According to the thresholds suggested by Hair *et al.* (2017a), the VAF results indicate that IT represents a substantial mediating mechanism, whereas CDM exhibits a moderate mediating effect. These findings demonstrate that DTT enhances OME both directly and indirectly through increased IT and CDM, with IT representing the stronger mediating mechanism.

Table 10. Indirect effects

Construct	Indirect effect (β)	*T* value	*p* value	Mediation type
DTT → IT → OME	0.749	9.805	0.000	Partial mediation
DTT → CDM → OME	0.202	3.321	0.001	Partial mediation

Note(s): Collaborative Decision-Making, Digital Twin Technology, Information Transparency, Organizational Digital Readiness, Operations Management Efficiency

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5.5 Necessary condition analysis (NCA)

NCA was conducted as a complementary analysis to PLS-SEM because the two approaches address different aspects of the relationships among the constructs. Whereas PLS-SEM estimates the average contribution of predictors to an outcome, NCA examines whether a minimum level of a condition is required for a specified level of the outcome to be attainable (Dul, 2016; Richter *et al.*, 2020). This distinction is particularly relevant to the present study because DTT, IT, CDM, and ODR may contribute to OME while also functioning as enabling conditions for achieving higher levels of operational efficiency.

The effect size (*d*) of the latent construct scores, along with their *p*-values, was computed and tested following the recommended approach by Dul (2016). According to this approach, three conditions must be satisfied for a variable to be considered a necessary condition: *(i)* theoretical justification, *(ii)* effect size (d) greater than 0.1, and *(iii)* statistical significance ($p < 0.05$). The results presented in Table 11 show that all exogenous variables exhibit significant *p*-values. The NCA findings indicate that all latent constructs, including CDM, DTT, IT, ODR, and the interaction term (ODR × DTT), are meaningful ($d \geq 0.1$) and significant ($p < 0.05$) necessary conditions for achieving OME.

Figure 4 illustrates the scatter plots between each predictor (i.e. CDM, DTT, IT, ODR, and ODR × DTT) and the outcome variable OME. The bottleneck analysis (Table 12) further provides a detailed assessment of each necessary condition. To achieve the desired level of the dependent variable (OME), progressively higher levels of the predictor variables become necessary, indicating that these factors function as enabling conditions for superior operational performance.

In Table 12, "NN" denotes "Not Necessary," indicating that no minimum threshold level of the condition is required to achieve the corresponding outcome level. This result suggests that at lower levels of OME, specific antecedent conditions are not mandatory. However, as the desired level of OME increases, progressively higher levels of the necessary conditions become required.

The bottleneck analysis further provides insights into the minimum levels of each condition required to achieve specific levels of OME. The results indicate that at lower levels of OME (0–10%), no specific conditions are required. However, as the desired level of OME increases, minimum thresholds for each variable become necessary.

For instance, to achieve 50% of OME, minimum levels of CDM $_{(0.922)}$, DTT $_{(1.397)}$, IT $_{(0.543)}$, ODR $_{(1.814)}$, and the interaction term (1.473) are required. At higher performance levels (e.g. 90%), the required levels increase further, particularly for DTT $_{(1.308)}$ and IT $_{(1.329)}$, highlighting their critical role. At the maximum level of OME (100%), all conditions must be present at relatively high levels, especially the interaction term (ODR × DTT$=_{5.228}$), IT $_{(2.512)}$, and CDM $_{(2.486)}$.

Table 11. NCA effect size

Construct	Effect sizes (*d*) CE-FDH (OME)	*p*-value
LV scores - CDM	0.363	0.000
LV scores - DTT	0.274	0.000
LV scores - IT	0.395	0.000
LV scores - ODR	0.231	0.000
LV scores - ODR x DTT	0.109	0.000

Note(s): Effect size of ($d = 0 < d < 0.1$) signifies "weak effect size", ($d = 0.1 < d < 0.3$) represents "moderate effect size", ($d = 0.3 < d < 0.5$) shows "large effect size" and ($d = 0.5 < d < 01$) indicates "very large effect size"

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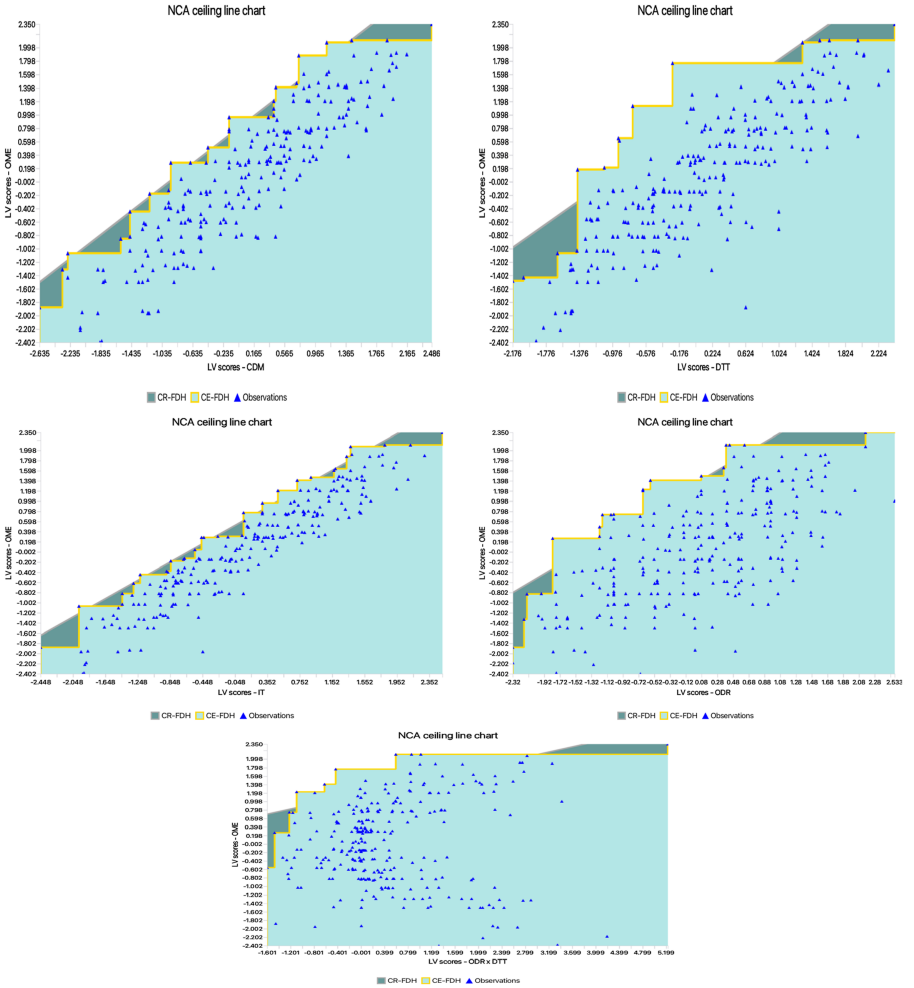


Figure 4. Scatter plots of the predictor variables versus OME. **Source(s):** Authors' own creation/work

Beyond identifying necessary conditions, Figure 4 also offers important insights into how the examined factors constrain OME. The empty upper-left regions across the scatter plots indicate that high levels of OME are rarely achieved when IT, CDM, DTT, or ODR remain at low levels, suggesting that each factor must satisfy a minimum threshold before superior operational outcomes become attainable. Among these conditions, IT exhibits the most pronounced ceiling line, indicating that IT is the strongest necessary condition for achieving high levels of operational efficiency. This finding is consistent with the PLS-SEM results, where IT also demonstrated the largest positive effect on OME, reinforcing its role as both a performance-enhancing mechanism and a prerequisite for superior operational outcomes in construction projects.

The scatter plots for DTT and CDM further suggest that their contributions operate through threshold effects rather than linear improvements alone. Specifically, insufficient digital twin capabilities or CDM constrain operational performance, regardless of improvements in other organizational factors. Likewise, the interaction term (ODR × DTT) exhibits an increasingly

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Table 12. Bottleneck analysis (Robustness test)

Percentile	LV scores - CDM	LV scores - DTT	LV scores - IT	LV scores - ODR	LV scores - ODR x DTT
0%	NN	NN	NN	NN	NN
10%	NN	NN	NN	NN	NN
20%	2.336	2.045	1.976	2.177	NN
30%	1.575	1.397	1.440	2.140	NN
40%	1.454	1.397	1.220	1.814	1.473
50%	0.922	1.397	0.543	1.814	1.473
60%	0.436	0.906	0.056	1.218	1.222
70%	0.160	0.733	0.289	0.666	1.097
80%	0.450	0.255	0.719	0.572	0.622
90%	0.749	1.308	1.329	0.390	0.592
100%	2.486	2.418	2.512	2.163	5.228

Source(s): Authors' own creation/work

restrictive bottleneck at higher levels of OME, indicating that organizations seeking exceptional operational performance must combine strong digital twin capabilities with sufficient ODR. Possessing high levels of one without the other is unlikely to generate maximum efficiency gains.

The NCA results complement the PLS-SEM findings by providing a different perspective on the role of the predictor variables. While PLS-SEM demonstrates that IT, CDM, DTT, and ODR contribute positively to OME, NCA shows that these factors also function as necessary conditions at higher levels of performance. The convergence of sufficiency-based (PLS-SEM) and necessity-based (NCA) findings strengthens confidence in the robustness of the results. In particular, IT emerges as the most influential construct across both analytical approaches, suggesting that transparent information flows represent both the primary mechanism and a critical prerequisite for achieving superior operational efficiency in digitally enabled construction environments.

6. Discussion and implications

This study examined the impact of DTT on OME, along with the mediating roles of IT and CDM and the moderating role of ODR. Overall, the findings provide support for the majority of the proposed relationships. However, while the moderating effect of ODR was statistically significant, its direction was opposite to that originally hypothesized. Therefore, the results provide evidence for the existence of a moderation effect but do not support the hypothesized positive moderating relationship. In a dual-method design, the research used PLS-SEM to evaluate the hypothesized relationships among these constructs and NCA to discover the conditions required to achieve efficacious results in the construction industry. These analyses provide complementary evidence: PLS-SEM identifies the strength and direction of the relationships among the focal constructs, whereas NCA identifies whether minimum levels of particular capabilities are required before higher levels of OME can be achieved.

H1 indicates that DTT has a significant positive effect on OME. This suggests that the adoption of DTT enhances operational performance through improved monitoring, simulation, and process optimization. This finding is consistent with prior studies highlighting the role of DT systems in improving operational efficiency (Alguirat *et al.*, 2025; Fantozzi *et al.*, 2025). However, the significance of this finding is particularly pronounced in construction because operational activities occur within constantly changing physical environments rather than stable production systems. Unlike manufacturing settings where workflows are repetitive and predictable, construction projects involve evolving site

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conditions, shifting resource requirements, and temporary stakeholder configurations. Under these circumstances, DTT creates value by continuously synchronizing physical and digital environments, enabling project teams to identify emerging issues, anticipate disruptions, and implement corrective actions before operational inefficiencies escalate.

H2 and H3 show that DTT significantly enhances both IT and CDM. This indicates that DTT improves real-time data visibility while also facilitating better coordination and interaction among stakeholders (Patil *et al.*, 2024). These results align with prior research suggesting that DTT enhances information sharing and collaborative processes within organizations (Lee *et al.*, 2025). The construction context provides an important explanation for these relationships. Construction projects are typically executed through temporary coalitions of contractors, subcontractors, consultants, and clients who frequently operate using separate information systems and organizational objectives. Consequently, information fragmentation and coordination failures represent persistent industry challenges. DTT addresses these challenges by creating a shared digital representation of project activities that enables stakeholders to access consistent, real-time information, thereby improving both transparency and collaboration.

Moreover, the findings demonstrate that IT represents the strongest mechanism linking DTT with operational efficiency. This finding is consistent with previous studies emphasizing the importance of timely, accurate, and transparent information for improving organizational performance (Liew, 2019). However, the magnitude of this relationship should be interpreted with appropriate caution. The exceptionally large coefficient observed for the IT → OME relationship ($\beta = 0.925$) indicates a very strong statistical association in the present sample, but should not be interpreted as evidence that IT alone determines operational efficiency. Although the validity and collinearity diagnostics support the adequacy of the measurement model, the possibility that conceptual proximity between information-related capabilities and operational outcomes contributed to the magnitude of the relationship cannot be completely excluded. Consequently, the findings should be interpreted as evidence of a strong positive relationship rather than as indicating an almost deterministic influence of IT on operational efficiency. The unusually large coefficient therefore represents an important empirical finding that warrants further validation using more technology-neutral and objective measures of operational performance.

The findings also demonstrate that CDM positively influences OME, although the magnitude of this relationship is considerably smaller. The contrast between the two mediation pathways suggests that, within construction projects, transparency is likely to precede collaboration. Construction projects typically involve multiple contractors, consultants, subcontractors, and clients operating across organizational boundaries. Without reliable and transparent information, CDM becomes substantially more difficult because project participants lack a common understanding of project status, resource availability, and emerging risks. Accordingly, DTT appears to create operational value by reducing information asymmetry and improving information visibility, which in turn provides project participants with a shared basis for coordinating activities, resolving emerging issues, and making timely operational decisions. This interpretation should nevertheless be understood as an empirical pattern within the present sample rather than as evidence that transparency is universally more important than collaboration across all construction contexts.

Contrary to the proposed hypothesis, ODR exhibited a statistically significant but negative moderating effect on the relationship between DTT and OME ($\beta = -0.056$, $p = 0.002$). Accordingly, H6 was not supported in the hypothesized direction. Although unexpected, the effect size remained relatively small ($f^2 = 0.037$), suggesting that digital readiness represents a modest boundary condition rather than a dominant determinant of operational efficiency.

One plausible explanation is that organizations possessing mature digital infrastructures have already realized many of the operational improvements associated with digitalization. Consequently, the incremental benefits attributable specifically to DTT become comparatively smaller because existing digital capabilities already perform many

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information-processing and coordination functions. Conversely, organizations with lower levels of digital readiness may experience more substantial efficiency improvements following DTT adoption because digital twins address previously unmet operational needs.

This finding should therefore not be interpreted as suggesting that ODR is detrimental. Rather, it indicates that the marginal contribution of an additional digital technology depends upon the organization's existing digital capability base. Future research should investigate this unexpected relationship across different organizational contexts and examine whether the negative moderation effect is replicated using longitudinal or objective operational-performance data.

This interpretation is consistent with the competing explanation introduced in the theoretical development, which recognizes that digital maturity may generate both enabling and constraining effects. At high levels of digital maturity, resource redundancy, technological complexity, established routines, and organizational inertia may reduce the marginal contribution of an additional digital layer. The negative moderation should therefore not be interpreted as evidence that ODR itself is detrimental. Rather, it suggests that the incremental value of DTT may depend on the organization's existing digital capability base and the degree to which DTT provides capabilities that are genuinely additional to those already available.

The unexpected direction of the moderation effect also suggests that the relationship between ODR and digital twin effectiveness may be more complex than assumed in the original theoretical model. Rather than uniformly strengthening technology-performance relationships, digital readiness may alter the mechanisms through which value is created, potentially shifting the relative importance of IT, collaboration, or other organizational capabilities. This finding provides an important boundary condition for ROT: complementary resources may enhance the value of a focal technology when they fill capability gaps, but their marginal contribution may become smaller when organizations already possess extensive overlapping digital resources.

The exceptionally strong influence of IT compared with CDM suggests that construction organizations may derive greater operational benefits from reducing information fragmentation than from increasing collaborative interactions alone. Unlike manufacturing environments where processes are relatively standardized, construction projects often involve temporary teams, geographically dispersed activities, and multiple subcontractors. Under such conditions, transparency may serve as the foundational capability upon which effective collaboration depends. Consequently, DTT appears to create value primarily by establishing a shared and reliable information environment rather than merely increasing stakeholder interaction. Given the exceptionally large IT → OME coefficient, however, this interpretation should be treated as an empirical indication of relative importance rather than as evidence of a deterministic hierarchy between IT and collaboration.

The NCA findings provide an additional perspective on this resource-orchestration process. While the PLS-SEM results identify the relationships through which DTT and its associated mechanisms contribute to OME, the NCA results indicate that the focal capabilities also function as necessary conditions for achieving higher levels of operational efficiency. From an ROT perspective, this suggests that resource structuring represents a fundamental prerequisite for subsequent resource bundling and leveraging. In practical terms, organizations cannot fully exploit DTT-enabled collaboration and decision-making when the underlying information, technological, and organizational foundations remain below the necessary threshold. The NCA results therefore complement the mediation findings by indicating that some minimum capability level must be established before the broader benefits of digital resource orchestration can be realized.

6.1 Theoretical implications

A primary theoretical contribution of this study is the extension of ROT by explicitly demonstrating how digital twin resources are transformed into operational value through distinct orchestration mechanisms. Specifically, the findings show that DTT functions as a

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strategic digital resource, IT represents the structuring of information resources, and CDM reflects the bundling and leveraging of those resources to improve operational outcomes. By empirically validating these relationships, the study moves beyond generic discussions of digital technologies and provides a more precise explanation of how resource orchestration unfolds within construction environments.

The findings further suggest that resource orchestration may involve a sequential or hierarchical logic in digitally enabled construction environments. The substantially stronger relationship between IT and OME compared with CDM indicates that information visibility may provide an important foundation for subsequent collaborative processes. This extends ROT by suggesting that digital resources may not contribute independently or uniformly to organizational outcomes; rather, some resources and capabilities may provide the conditions under which subsequent orchestration activities become effective.

The study further contributes to the construction digitalization literature by distinguishing DTT from broader digital technologies and traditional BIM-based approaches. The findings suggest that the unique value of DTT lies not merely in digital representation but in its ability to facilitate real-time, bidirectional synchronization between physical and digital project environments. This distinction is particularly important in construction settings characterized by evolving site conditions, temporary organizational structures, and fragmented stakeholder networks. By focusing on these construction-specific realities, the study provides a more contextually grounded explanation of digital twin value creation.

Another important theoretical contribution is the identification of ODR as a boundary condition of resource orchestration effectiveness. Unlike prior assumptions that digital readiness uniformly strengthens technology-performance relationships, the findings reveal a significant but negative moderating effect. This result suggests that the value generated by DTT depends not only on the availability of organizational capabilities but also on the extent to which existing digital resources may substitute for or overlap with the functions provided by digital twins. Consequently, the study extends ROT by demonstrating that contextual organizational conditions may not always amplify resource value; under certain circumstances, they may reduce the marginal contribution of additional digital resources. This finding introduces a digital-readiness paradox into the resource-orchestration perspective: the same digital capabilities that enable organizations to exploit new technologies may, at sufficiently high levels, reduce the incremental value of additional technologies through redundancy, integration complexity, or established organizational routines.

The NCA findings provide a further theoretical contribution by identifying the importance of necessary conditions within the resource-orchestration process. The results suggest that certain digital and organizational capabilities must reach minimum levels before higher levels of OME can be achieved. This finding complements the conventional emphasis of ROT on structuring, bundling, and leveraging by highlighting a bottleneck logic: resource structuring establishes the minimum foundation upon which subsequent resource bundling and leveraging can operate effectively. Thus, the orchestration process is not simply a matter of possessing resources or combining them; it also depends on whether foundational capabilities have reached sufficient levels to permit higher-order value creation.

Finally, the findings also contribute to the growing literature on digital transformation by demonstrating that statistically significant relationships should not necessarily be interpreted as practically dominant relationships. The unusually large coefficients observed for IT and the high explanatory power of the model warrant cautious interpretation and highlight the importance of continued refinement of construct measurement within digital transformation research. The magnitude of these estimates should therefore be viewed as evidence of strong relationships within the present empirical setting rather than as evidence of universal or deterministic effects. Future studies employing more technology-neutral operational performance measures will help determine the extent to which these relationships generalize across different organizational settings.

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6.2 Managerial implications

The findings suggest that construction firms should adopt a structured approach to digital transformation. Rather than investing simultaneously in multiple digital technologies, managers should prioritize establishing reliable information infrastructures capable of generating accurate, timely, and transparent project information. Once these information systems have been successfully implemented, organizations should focus on strengthening CDM processes that leverage the improved information environment. The findings indicate that collaboration is likely to yield greater operational benefits when supported by transparent and reliable information, rather than when implemented as an isolated organizational initiative.

The combined PLS-SEM and NCA findings suggest that digital transformation should be approached as a capability-building sequence rather than as an immediate technology investment. Managers should first assess whether the organization possesses the minimum technological, informational, and organizational capabilities required to exploit DTT. Where foundational capabilities are insufficient, investments in data infrastructure, interoperability, employee digital skills, and information governance should precede full-scale DTT deployment. Once these foundational conditions are established, organizations can progressively integrate real-time DTT capabilities and subsequently embed them within collaborative planning and decision-making processes.

This sequencing is important because the NCA results indicate that the focal capabilities function as necessary conditions for achieving higher levels of OME. Accordingly, managers should identify and address capability bottlenecks before investing heavily in advanced digital twin applications. In practical terms, the implementation sequence should begin with establishing adequate digital and information foundations, proceed to ensuring reliable real-time information visibility, and then focus on using that information to support CDM and operational coordination.

For project managers, DTT can be used to create real-time dashboards that track construction progress, equipment utilization, safety incidents, and resource allocation across project stages. For site engineers, digital twins can support rapid identification of deviations between planned and actual site conditions, reducing delays and rework. The findings also suggest that DTT investments should be aligned with existing BIM capabilities. Rather than replacing BIM, digital twins should be implemented as an extension that enables continuous real-time synchronization between physical assets and digital models. However, organizations should evaluate the incremental functionality provided by DTT rather than assuming that greater digital maturity automatically produces greater returns. Since construction projects involve temporary teams and multiple contractors, managers should focus on creating a shared information environment that reduces owner–contractor information asymmetries and improves coordination across project participants.

The negative moderating effect of ODR provides an important qualification to digital investment decisions. The results indicate that organizations with lower or developing levels of digital readiness may obtain larger marginal operational gains from DTT because the technology addresses more substantial existing capability gaps. In contrast, highly digitally mature organizations may experience diminishing marginal returns when additional DTT functionality overlaps with existing information systems, analytics platforms, monitoring technologies, or coordination routines. Managers in digitally mature firms should therefore prioritize interoperability, system integration, process simplification, and elimination of redundant digital capabilities rather than simply adding another digital layer.

Accordingly, digital readiness should be treated as a basis for determining the appropriate form and scale of DTT investment rather than as a target that should be maximized irrespective of organizational circumstances. Firms with developing digital infrastructures should prioritize foundational capability development and targeted DTT applications that address clearly identified operational gaps. Firms with more mature digital ecosystems should instead conduct capability audits, identify functional overlaps, and ensure that DTT complements

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rather than duplicates existing systems. This approach can help organizations avoid unnecessary technological complexity while preserving the operational benefits of digital twins.

7. Limitations and future research directions

This study has certain limitations that also provide opportunities for future research. The use of a cross-sectional survey design and perceptual data may introduce response bias and limits the ability to establish how the relationships among DTT, IT, CDM, ODR, and OME evolve over time. Although the hypothesized relationships are theoretically grounded and statistically significant, the cross-sectional design does not permit conclusions about temporal ordering or changes in the strength of these relationships as organizations gain experience with DTT. Future research could therefore employ longitudinal designs to examine how the operational benefits of DTT develop across different stages of implementation and whether the role of IT, CDM, and ODR changes as organizations become more experienced with digital twin technologies.

The study is also limited to construction firms in Shandong Province, China, which may constrain the generalizability of the findings to other geographical, institutional, and organizational contexts. Construction organizations operating in different countries may differ in their technological infrastructure, regulatory environments, project delivery practices, organizational cultures, and levels of digital maturity. The observed relationships, particularly the negative moderating effect of ODR, should therefore be interpreted within the empirical context of the study rather than assumed to apply universally. Future research should replicate the model across different countries, regions, and construction markets to determine whether the proposed relationships remain stable across institutional and technological environments. Comparative studies could also examine whether the role of ODR differs between developed and emerging digital ecosystems.

The study also does not incorporate respondent demographic and organizational characteristics as formal control variables in the structural model. Although such information was collected, the analysis focuses on the theoretical relationships among the principal constructs. Consequently, the estimated relationships should be interpreted as associations among the focal constructs rather than as effects adjusted for all potential sources of organizational heterogeneity. Future studies could incorporate relevant controls such as firm size, project size, project complexity, organizational age, respondent experience, project type, previous DTT experience, and existing levels of digital maturity. Such extensions would provide a more comprehensive assessment of whether the observed relationships remain robust after accounting for organizational and project-level differences.

The exceptionally large coefficient between IT and OME and the very high explanatory power of the model also warrant cautious interpretation. Although the measurement model satisfies the reported reliability, validity, and collinearity criteria, the magnitude of the IT → OME relationship may partly reflect the conceptual proximity between information-related capabilities and perceived operational efficiency. In addition, the use of perceptual measures collected from the same respondents may increase the possibility of common-method effects. The coefficient should therefore be interpreted as evidence of a very strong empirical association within the present sample rather than as evidence that IT almost deterministically determines OME. Future research should cross-validate this relationship using independent data sources, longitudinal observations, and objective indicators of operational performance.

The measurement of OME represents another area requiring further validation. Several OME indicators capture operational improvements that may be closely associated with the benefits expected from digital technologies. Although this approach is consistent with the study's objective of examining technology-enabled operational efficiency, it may create some conceptual overlap between the digital capabilities examined as predictors and the operational outcomes used as dependent measures. Future studies could strengthen construct separation by

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incorporating more technology-neutral measures of OME, such as schedule adherence, productivity, cost variance, resource utilization, rework rates, defect rates, equipment downtime, and operational response time. Combining perceptual measures with objective project-level indicators would provide stronger evidence regarding the extent to which DTT contributes to measurable improvements in construction operations.

The negative moderating effect of ODR also requires further investigation. The present study hypothesized that higher digital readiness would strengthen the positive relationship between DTT and OME, but the empirical results indicate a statistically significant dampening effect. The relatively small interaction effect suggests that ODR is not a dominant determinant of the DTT–OME relationship; nevertheless, the finding raises an important theoretical question regarding the possibility of diminishing marginal returns from additional digital technologies in highly digitally mature organizations. Organizations with extensive digital infrastructures may already possess information systems, analytics capabilities, monitoring technologies, and technology-enabled coordination routines that overlap with some DTT functionalities. Resource redundancy, integration complexity, established routines, and organizational inertia may therefore reduce the incremental operational contribution of DTT when baseline digital capabilities are already highly developed. Future research should investigate these mechanisms directly rather than treating the negative moderation as a single aggregate effect. Nonlinear models, multi-group comparisons, and longitudinal studies could determine whether the relationship between ODR and DTT value follows a threshold or diminishing-returns pattern.

The unexpected moderation result also suggests that digital readiness may operate differently across stages of organizational digital transformation. At relatively low levels of readiness, additional digital capabilities may enable organizations to overcome substantial information and coordination deficiencies. At higher levels, however, additional technologies may generate smaller incremental benefits because existing systems already address many of the underlying operational requirements. Future research could therefore examine whether ODR has an inverted-U or other nonlinear relationship with the marginal operational value of DTT. Such research would help clarify whether the negative interaction observed in this study represents a general boundary condition or a characteristic of the particular organizational context examined.

The findings also indicate that IT and CDM play different roles in the DTT–OME relationship, with IT exhibiting a substantially stronger mediating effect than CDM. This difference should not be interpreted as establishing a universal hierarchy in which IT is always more important than CDM. The relative strength of these mechanisms may depend on project complexity, stakeholder fragmentation, procurement arrangements, contractual relationships, and the maturity of interorganizational collaboration. Future studies could therefore investigate whether the relative importance of IT and CDM changes across project types, project delivery methods, and stages of the construction lifecycle. Qualitative and mixed-method research could further explain how transparent information is converted into collaborative actions during actual construction operations.

The NCA findings also provide an important direction for future research. The identification of necessary conditions suggests that achieving higher levels of OME may require minimum levels of the underlying digital and organizational capabilities. However, the cross-sectional design does not establish whether these thresholds remain stable across different projects, organizational settings, or stages of digital transformation. Future research could use longitudinal and comparative designs to examine whether the identified bottlenecks change as organizations develop their digital capabilities. Such studies could further investigate the sequence through which resources are structured, bundled, and leveraged, thereby providing stronger evidence for the proposed integration of NCA with ROT.

The present study focuses on DTT, IT, CDM, ODR, and OME and therefore does not capture all organizational and environmental factors that may influence digital transformation outcomes in construction. Future research could extend the model by examining additional

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contextual conditions, including organizational agility, digital leadership, technological turbulence, organizational resilience, interoperability capability, and project complexity. These factors may help explain why similar DTT investments generate different operational outcomes across organizations and projects.

Future research could also examine more complex conditional mechanisms. For example, ODR may moderate not only the overall DTT–OME relationship but also the individual pathways through which DTT enhances IT and CDM or through which these mechanisms contribute to OME. Moderated-mediation models could therefore provide a more detailed understanding of whether digital readiness changes the strength of the individual value-creation mechanisms rather than simply altering the overall relationship between DTT and OME.

Finally, replication using independent samples and alternative methodological approaches would strengthen confidence in the robustness of the findings. Case studies could provide detailed evidence of how DTT resources are structured and integrated into construction operations, while longitudinal and multi-source studies could test whether the strong relationships identified in the present analysis persist when technology adoption and operational performance are measured independently over time. Such research would be particularly valuable for determining whether the high explanatory power and strong IT–OME relationship observed in this study are characteristic of construction organizations generally or are partly attributable to the measurement and sampling context of the present study.

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